

ProteinChip[®] System Starter Kit

Instruction Manual

Catalog #C70-00068

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Part I: ProteinChip® Starter Kit Introduction

Chapter 1 ProteinChip Starter Kit

1.1 Product Description

The ProteinChip SELDI system uses surface-enhanced laser desorption/ionization (SELDI) technology to capture, detect, and calculate the mass of peptides and proteins in complex biological samples. The system integrates a ProteinChip reader with data analysis software to analyze proteins selectively captured on ProteinChip arrays.

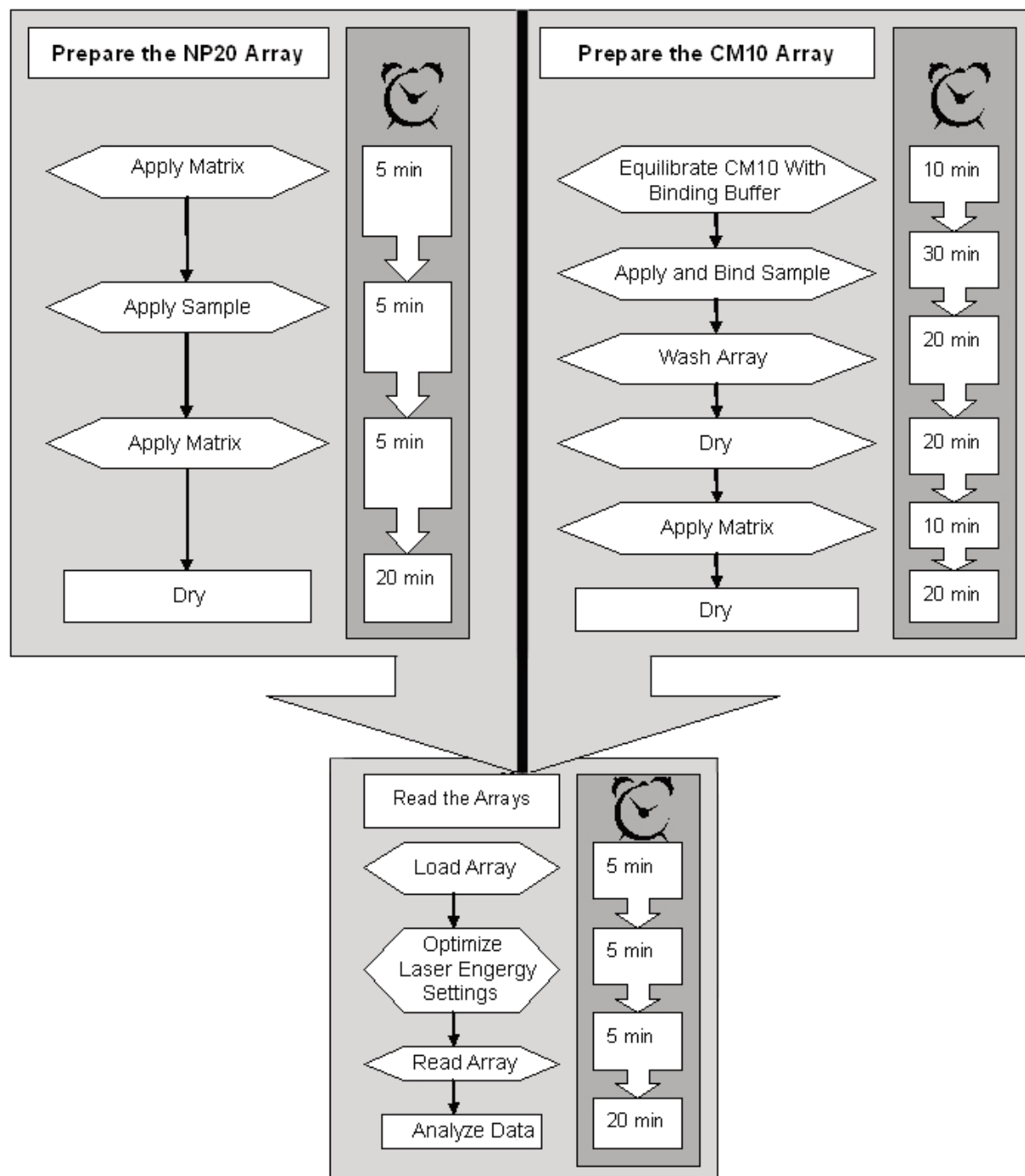
The ProteinChip starter kit provides the reagents, arrays, samples, instrument protocols, and instructions required to prepare and analyze ProteinChip arrays using the ProteinChip SELDI system. Use of this kit complements the ProteinChip SELDI system basic course; it demonstrates the basic steps involved in preparing an array, loading arrays into the ProteinChip reader, protocol setup, and basic software functions such as peak viewing, peak labeling, and mass calibration.

This instruction manual is divided into two parts that describe the protocols used with this kit (see table below).

Instruction manual used with the ProteinChip starter kit.

Name	Description	Arrays Used
Part II: Basic System Operation and Data Collection	Demonstrates ProteinChip array handling and loading, protocol setup, and basic software functions such as protocol optimization, peak labeling, and mass calibration	ProteinChip peptide standard array
Part III: Array Preparation and Basic Data Analysis	Demonstrates the basic ProteinChip array preparation and analysis workflow and allows you to evaluate your technique against premade ProteinChip arrays	ProteinChip NP20 array ProteinChip premade NP20 array ProteinChip CM10 array ProteinChip premade CM10 array

The following workflows summarize the steps to follow for each analysis method.



1.2 Kit Components and Storage Conditions

Component	Qty	Description	Storage Conditions
Reference Samples			
Human serum, 0.5 ml	1	Pooled human serum	–20°C to –80°C
<i>Escherichia coli</i> lysate	1	Lyophilized cell lysate	4°C
ProteinChip Arrays			
ProteinChip peptide standard array	1	ProteinChip NP20 array prepared with a mix of peptide standards and matrix	Ambient to 4°C
ProteinChip CM10 arrays	3	Weak cation exchange surface; allows capture of positive charges, such as lysine, arginine, and histidine residues	Ambient to 4°C
ProteinChip NP20 arrays	3	Silicon dioxide surface; allows capture through serine, threonine, or lysine residues	Ambient to 4°C
ProteinChip premade CM10 array	1	ProteinChip CM10 array prepared with matrix and serum	–20°C to –80°C
ProteinChip premade NP20 array	1	ProteinChip NP20 array prepared with matrix and <i>E. coli</i> and serum samples	–20°C to –80°C
Blank arrays	12	Blank ProteinChip arrays used to fill empty slots in the ProteinChip cassette	Ambient
ProteinChip Accessories			
ProteinChip cassettes	2	Used to load 12 arrays into the Enterprise Edition reader and to hold arrays in the ProteinChip bioprocessor	
ProteinChip cassette-compatible bioprocessor reservoir	1	Component of the bioprocessor	
Sealing foil	3	Used to seal bioprocessor during array preparation	
Reagents			
Sinapinic acid (SPA) matrix 5mg		Matrix used for general profiling	Ambient to 4°C
Acetonitrile, 2 ml	1	Solvent used to prepare matrix solution	Ambient to 4°C
Trifluoroacetic acid (TFA), 2 ml	1	Solvent used to prepare matrix solution	Ambient to 4°C
CM low-stringency buffer, 30 ml	1	Binding and washing buffer, pH 4	4°C
CD: Instructional video, protocols, and analysis spreadsheet	1		
Instruction manual	1		

1.3 Additional Requirements

- ProteinChip SELDI system, Enterprise or Personal Edition
- Proteomics-grade water
- Three 15 ml sterile tubes with caps
- Calibrated 0.001–1.0 ml pipets and sterile tips
- Ten sterile 1.5 ml microcentrifuge tubes (for preparing aliquots of samples)
- Benchtop vortexer
- Benchtop shaker (for example, MicroMix shaker)
- Benchtop microcentrifuge
- ProteinChip cassette-compatible bioprocessor (catalog #C50-30011)

1.4 General Recommendations

- Before performing any of the tasks described in this manual, view the video tutorial supplied with the kit (also available at bio-rad.com/proteinchip/starterkit). It contains an overview of the kit procedures and offers valuable advice about array handling and loading
- Store all arrays in their original packaging and do not open the packaging until ready to use
- Prepared arrays can be used multiple times, though the peak features will degrade over time. After opening, store arrays in a dry and dark location and their original packaging. Record the usage dates
- The ProteinChip premade arrays supplied with this kit have been prepared with a mixture of peptides and matrix. Avoid exposing these arrays to vibration, moisture, or light
- ProteinChip array spots contain chemically treated surfaces. Avoid contamination of these surfaces: always handle ProteinChip arrays by their edges, and never touch the spots
- When running single ProteinChip arrays on the ProteinChip instrument, Enterprise Edition, always fill empty locations in the cassette with blank, expired, or used arrays. Blank arrays are provided in this kit
- Use calibrated, small-volume pipets when applying sample and matrix to the arrays. For best results, use the smallest volume pipet appropriate (for example, use a 1 μ l or 2 μ l pipet to deposit 1 μ l matrix)
- To ensure best results, evaluate the performance of the ProteinChip system using the ProteinChip system check kit (catalog #C70-00081) or ProteinChip OQ kit (catalog #C70-00080), or as included in service contracts
- Protocol transfer from the CD to ProteinChip data manager software needs to occur only once
- The video tutorial included with this kit illustrates the use of ProteinChip data manager software version 3.0.7. Future releases of the software may include buttons or menu items not shown in this video. New features should not affect use of this kit

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Part II: Basic Operation and Data Collection

This protocol demonstrates the basic functions of the ProteinChip® reader and ProteinChip data manager software. It uses the ProteinChip peptide standard array, a ProteinChip NP20 array that has been prepared with a mix of peptide standards (Table 1) and matrix. The array is used with the instruction manual and video to demonstrate ProteinChip array handling and loading, protocol setup, and basic software functions such as protocol optimization, peak labeling, and mass calibration.

Table 1. Peptides included in the ProteinChip peptide standard array.

Peak	Name	Mass (Da)
1	Arg8-vasopressin	1,084.25
2	Somatostatin	1,637.90
3	Dynorphin A	2,147.50
4	ACTH	2,933.50
5	Beta endorphin	3,465.00
6	Arg-insulin	5,963.80
7	Cytochrome C	12,230.92

Chapter 2 Basic Operation and Data Collection

The following protocol demonstrates the basic functions of the ProteinChip reader and ProteinChip data manager software. It covers insertion of the ProteinChip array into the reader and explains the basic software functions required to obtain data from a ProteinChip array.

2.1 Import the Protocols

The ProteinChip reader uses data collection protocols to collect data from arrays. The protocols used with this kit are provided with the CD. Import them into the **Protocols** folder of ProteinChip data manager software.

1. Ensure that the ProteinChip reader is turned on and connected to the computer (this is usually done during installation).
2. Insert the ProteinChip Starter Kit CD and launch ProteinChip data manager software.
3. In the **Explorer** pane, select the **Protocols** folder. In the toolbar, select **Protocols > Import Protocol**. The **Choose File(s) to Import** dialog opens.
4. Select **Pep Std Array.ptx** and **Starter Kit.ptx** and click **Import**. Click **OK** at the prompt.

2.2 Load the Array and Gain Control of the Reader

2.2.1 Enterprise Edition

The Enterprise Edition reader uses a cassette to load up to 12 arrays at a time.

1. Remove the ProteinChip peptide standard array from its packaging and record its bar code number. Handling the array by its edges, gently push the array into slot #1 of the cassette, with spot H entering the cassette first.
2. Fill the empty slots of the cassette with blank arrays.
3. Place the cassette into the reader and close the lid. The reader moves the cassette into position while reading the bar codes of the arrays (blank arrays have no bar code). The arrays in the cassette are now in position to be loaded into the reader. Each array automatically loads one at a time when a protocol is run.
4. In the **Explorer** pane of ProteinChip data manager software, open the **Instruments** folder and select the reader from the list. The **Home** tab opens.
5. In the list of array bar codes, confirm the peptide standard array is in location #1.
6. Click the **Start** button  to gain control of the reader. The **Manual Mode** tab opens, indicating the arrays are in position and ready to be used.

2.2.2 Personal Edition

The Personal Edition reader analyzes one array at a time.

1. Remove the ProteinChip peptide standard array from its packaging and record its bar code number.
2. In the **Explorer** pane of ProteinChip data manager software, click the **Instruments** icon and select the reader from the list. The **Home** tab opens.
3. Click the **Start** button  to gain control of the reader.
4. Click the **Set Array Barcode** button . The **Set Array Barcode** dialog opens. Enter the array bar code and click **Set Barcode**.
5. Click the **Door Unlock** icon  in the toolbar to release the vacuum.
6. Once the blue light on the reader begins to flash, open the door and slide the array into the reader, with spot H entering first.
7. Check the gasket for any lint or dust that may prevent the reader from obtaining the necessary vacuum. If necessary, use a methanol- or ethanol-dampened, lint-free cloth to wipe the gasket clean.
8. Close the door and click the **Door Lock** icon . The reader prepares the array for analysis. The **Set Array Barcode** dialog opens and displays the current bar code. Click **Set Barcode**. The reader prepares the array for analysis.

2.3 Optimize the Laser Energy Setting (5.96 kD Peak, 100–200 μ A)

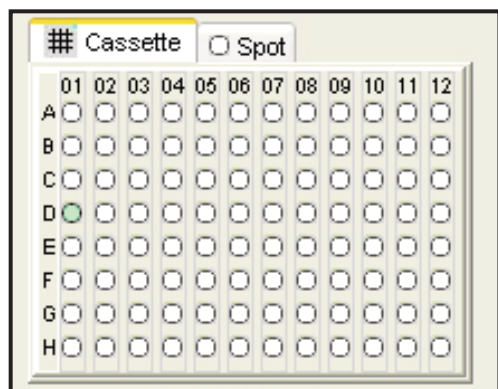
Data are collected from arrays using preset data collection protocols. For the best data quality, optimize the data collection protocols for each sample type. Optimize the laser energy settings for different sample types and to account for aging of reader components. Optimize the laser energy settings of a protocol each day, even when using the same sample type.

You imported a general protocol for analyzing the ProteinChip peptide standard array (Pep Std Array.ptx) in Section 2.1. Here, you optimize the laser energy such that the intensity for the peak at 5.96 kD is 100–200 μ A. An example of this procedure is provided in the instructional video.

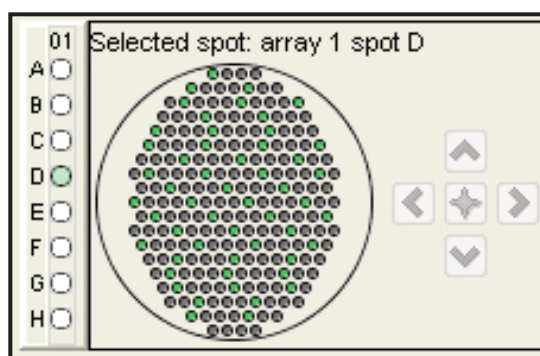
1. Open the **Protocol Mode** tab. In the **Acquisition Protocol** section, select the **Pep Std Array** protocol from the **Name** drop-down list.
2. Create the main folder for the starter kit data by clicking on the **Folder** button . The **Select Folder** dialog opens. Click and then right-click on the **Projects** folder and create a new project folder, **Starter Kit Data**.

3. Create a subfolder for the peptide standard array data by clicking again on the **Folder** button . The **Select Folder** dialog opens. Double-click on the **Projects** folder and then right-click on **Starter Kit Data**. Create a new project subfolder, **Pep Std Array**.
4. In Enterprise Edition software, the cassette map shows the distribution of arrays in the cassette. Columns 1–12 correspond to arrays 1–12. The peptide standard array is represented by column 1. Under column 1 (both software versions), select spot **D**. The spot turns green.

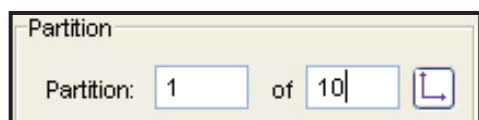
Enterprise Edition



Personal Edition



5. Set the partition to **1** of **10** and click the **Start Running Protocol** icon . The icon turns red, and the reader loads the array.



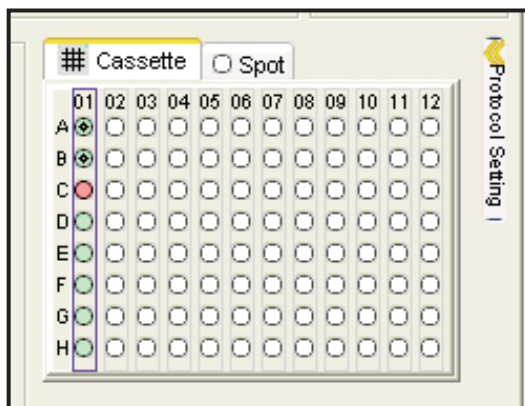
6. Once the spot is analyzed, the spectrum appears in the **Spectrum** section under the **Protocol Mode** tab (about 1 min).
7. Zoom in to the peak at 5.96 kD by clicking and dragging on either side of the peak, and then determine the intensity in μA .
8. Adjust the laser energy so that the peak intensity is 100–200 μA .

To change the laser setting, use the **Protocol Setting** tab. Adjust the laser energy using the **Data Shots(s) Energy** box. Adjust the warming shots using the **Warming Shots(s) Energy** box. Leave the **Add** box unchecked.

- If the intensity is less than 100 μA , increase the laser energy setting by 10% and increase the warming shot(s) energy to 10% higher than the data shot(s) energy
 - If the intensity is greater than 200 μA , decrease the laser energy setting by 10% and decrease the warming shot(s) energy to 10% higher than the data shot(s) energy
9. Run the protocol again using partition **2** of **10**. Changing the partitions allows the reader to collect data from a fresh location. Ensure spot **D** is selected and click the **Start Running Protocol** icon  to run the protocol with the new settings. Repeat steps 7–8, using a new partition each time (for example, the next run would be 3 of 10) until you obtain a peak intensity of 100–200 μA .
 10. Apply the optimized protocol to all spots on the array (see Section 2.4).

2.4 Collect Data and View and Label Peaks

1. Change the partitions back to **1** of **4**.
2. In the cassette map, select all 8 spots of the peptide standard array by clicking on each spot. The spots turn green. Click the **Start Running Protocol** icon .
3. The reader analyzes each spot (2 min each). The following icons indicate the status of each spot:
 -  **B** Data have been collected (spot is green)
 -  **C** Data are being collected (spot is red)
 -  **D** Data are going to be collected (spot is green)



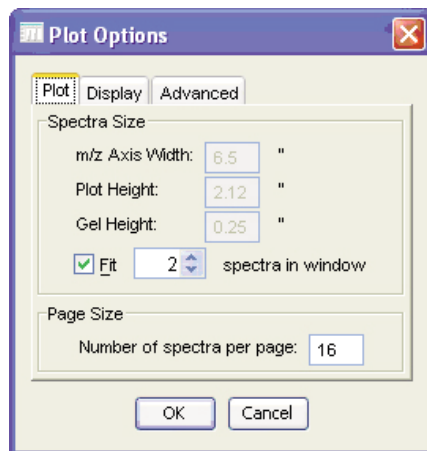
The status of each spot is also shown below the spectrum, along with information about the partition and protocol used.

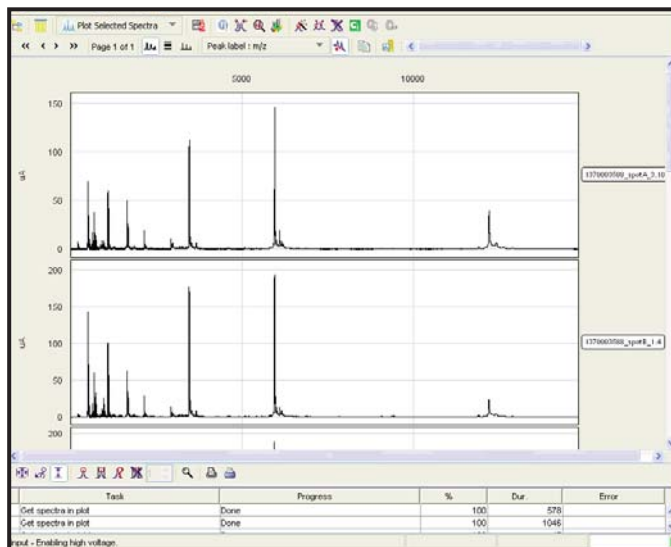
		Barcode	Array	Spot	Partition	Protocol
1	✓	1020100011	1	A	1 / 4	Pep Std Array
2	✓	1020100011	1	B	1 / 4	Pep Std Array
3		1020100011	1	C	1 / 4	Pep Std Array
4		1020100011	1	D	1 / 4	Pep Std Array
5		1020100011	1	E	1 / 4	Pep Std Array
6		1020100011	1	F	1 / 4	Pep Std Array
7		1020100011	1	G	1 / 4	Pep Std Array
8		1020100011	1	H	1 / 4	Pep Std Array

4. In the **Explorer** pane, locate and open the **Pep Std Array** folder. A list of spectra appears.

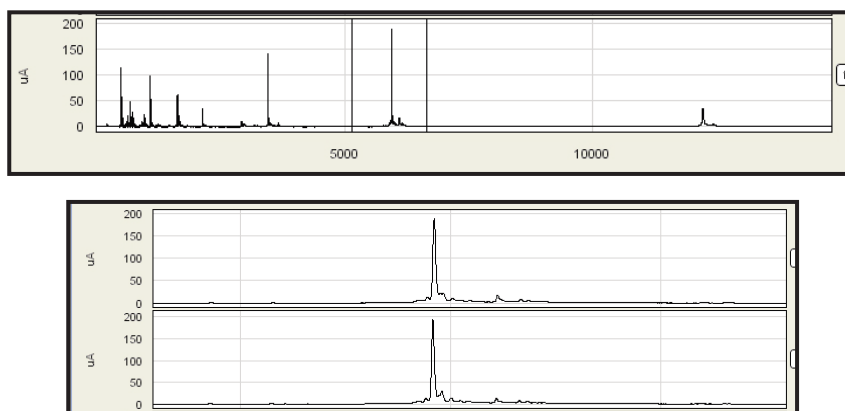
		Spectrum name	Date collected	...	...	...	Spot name	Frac
1		1170308422_spotD_1.10	2008-01-28 18:...	1			D	
2		1170308422_spotD_2.10	2008-01-28 18:...	1			D	
3		1170308422_spotA_1.4	2008-01-28 18:...	1			A	
4		1170308422_spotB_1.4	2008-01-28 18:...	1			B	
5		1170308422_spotC_1.4	2008-01-28 18:...	1			C	
6		1170308422_spotD_1.4	2008-01-28 18:...	1			D	
7		1170308422_spotE_1.4	2008-01-28 18:...	1			E	
8		1170308422_spotF_1.4	2008-01-28 18:...	1			F	
9		1170308422_spotG_1.4	2008-01-28 18:...	1			G	
10		1170308422_spotH_1.4	2008-01-28 18:...	1			H	

- Delete the spectra used for laser optimization. To find these spectra, sort the spectra by date collected. Spectra collected during optimization appear for spot D and with a **Date collected** value that precedes that for spot A. They may also have a spectrum name ending in "1.10" (1 of 10 partitions). Select and delete these spectra using the Delete key on your computer keyboard.
- Select the remaining 8 spectra from the **Plot Spectra** drop-down list and click the **Plot Selected Spectra** button  .
- Increase the viewing area by clicking the **Toggle Spectrum Table** icon  . Further adjust the view by clicking and dragging to enlarge or shrink it.
- Change the viewing area to display only 2 spectra by right-clicking on the spectra and selecting **Plot Options**. The **Plot Options** dialog appears. In the **spectra in window** drop-down list, change the number to **2**. Click **OK**. The window shows 2 spectra.

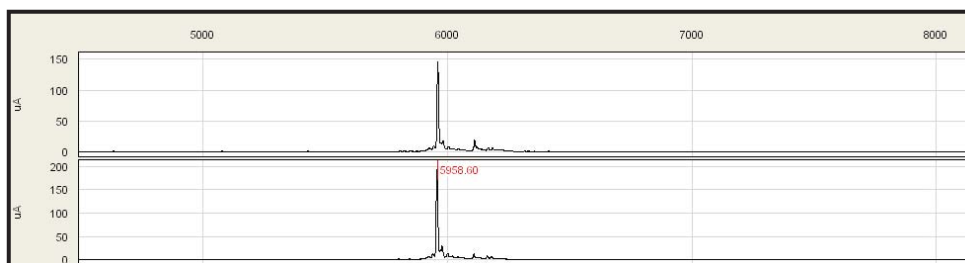




9. Scroll through the spectra and zoom in and out of certain regions. Click and drag to either side of a peak of interest, and use the slide bar to migrate through the spectra.
 - To zoom in, click on either side of the region of interest and use the slide bar to scroll through the region
 - To zoom out to the original view, click the **Reset Zoom** icon 
10. To label a peak in one spectrum, zoom into a single peak of interest. Click and drag to either side of the peak to zoom in.



11. Use the **Centroid** function to label the peak. Select the **Centroid** icon  and click on the peak of interest. The peak is labeled with the m/z value in all spectra that are open.



12. To label the same peak in multiple spectra, select several spectra then click the **Centroid All** icon .

Chapter 3

Mass Calibration

Mass calibration is the process by which the time-of-flight data obtained by mass spectrometry are converted to mass data. Data collected using standards (known peptides or proteins) are used to derive a calibration equation, which is then applied to the spectra of experimental samples. In the following protocol, you use the spectra obtained from the ProteinChip® peptide standard array to perform the mass calibration procedure.

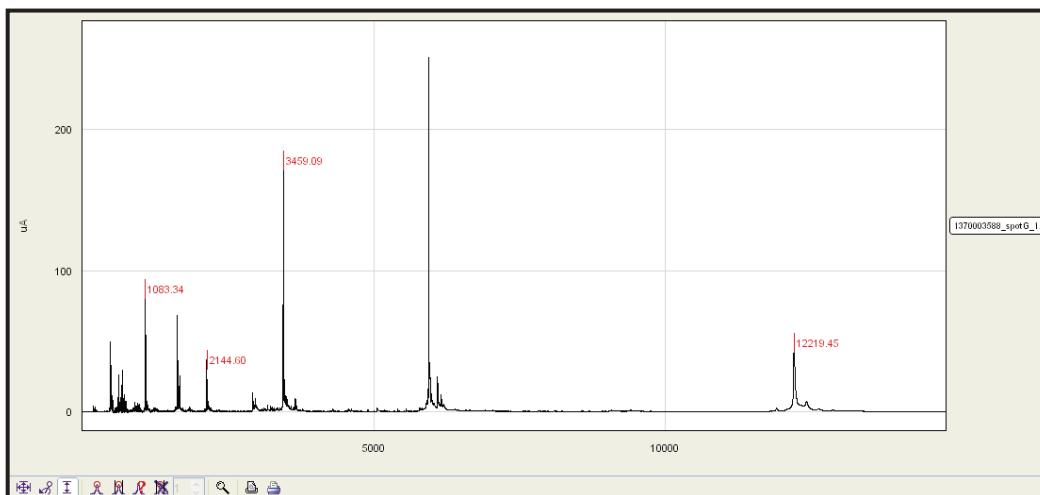
3.1 Perform Mass Calibration

1. Open any spectrum obtained using the peptide standard array, and label the 4 peaks indicated in the table below. Follow the instructions for peak labeling given in Section 2.4, Collect Data and View and Label Peaks.

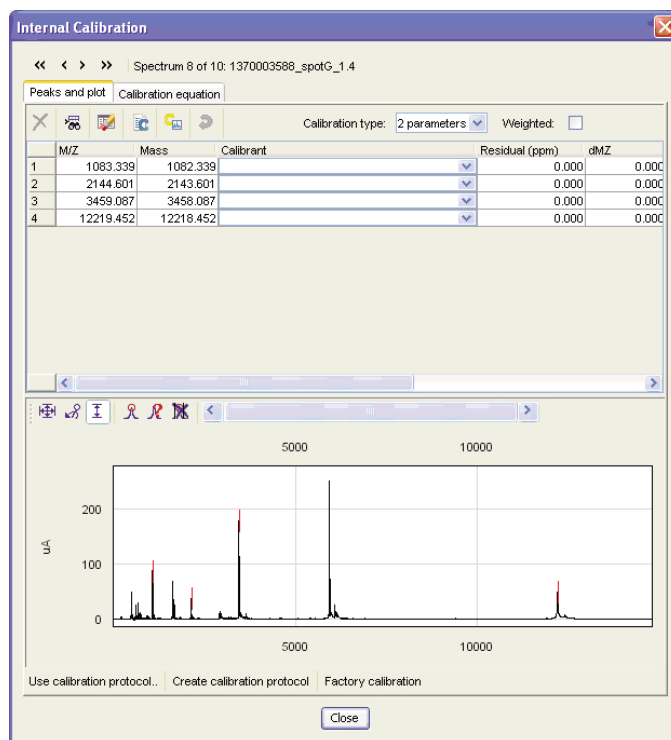
Note: Label each peak at its highest point; if not, the mass will be recorded incorrectly, reducing the accuracy of the calibration.

Label Peak for Calibration	Peak	Name	Molecular Mass (Da)
Yes	1	Arg8-vasopressin	1,084.25
No	2	Somatostatin	1,637.90
Yes	3	Dynorphin A	2,147.50
No	4	ACTH	2,933.50
Yes	5	Beta endorphin	3,465.00
No	6	Arg-insulin	5,963.80
Yes	7	Cytochrome C	12,230.92

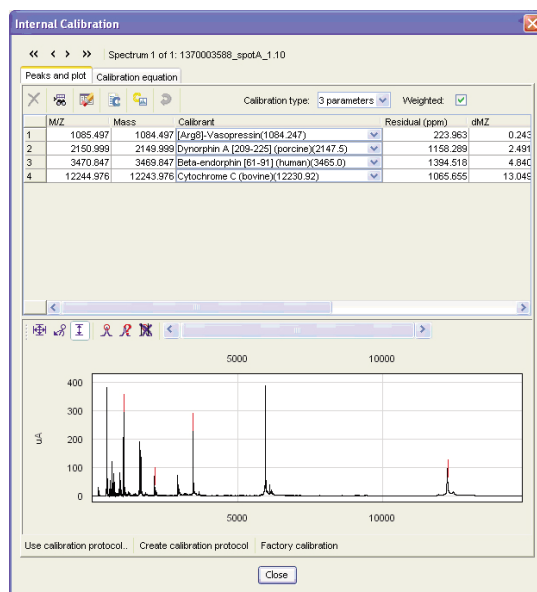
Mass standards included in the ProteinChip peptide standard array. Use the four peaks marked "yes" in the table above.



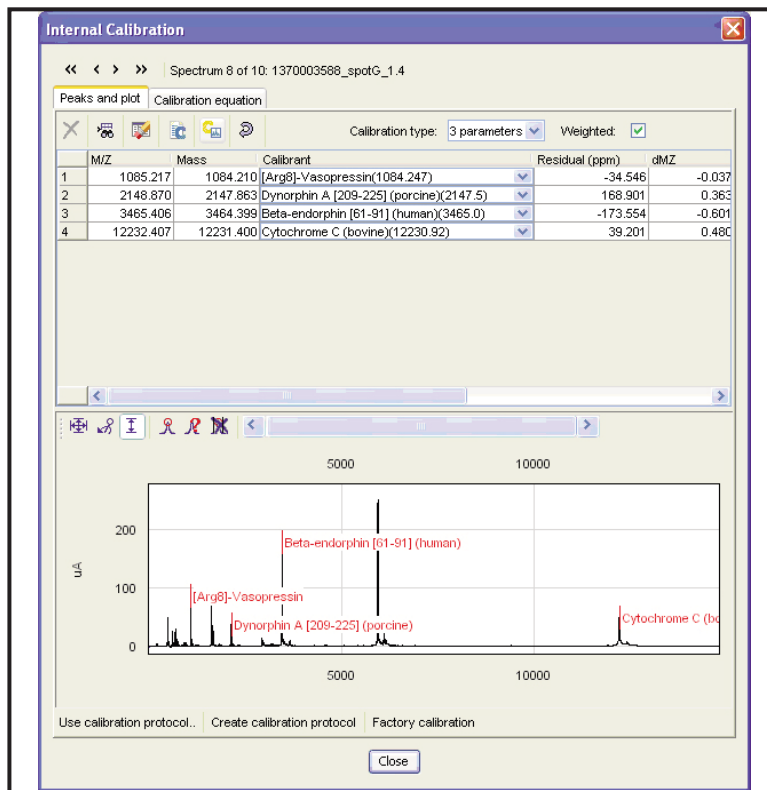
2. From the toolbar, select **Spectra > Calibrate > Internal Calibration** (or click the **Internal Calibration** icon ). The **Internal Calibration** dialog opens. The spectrum appears in the bottom half of the window with existing peak labels appearing in red. A table of calibrants and peaks appears in the top half of the window; the calibrant names are blank.



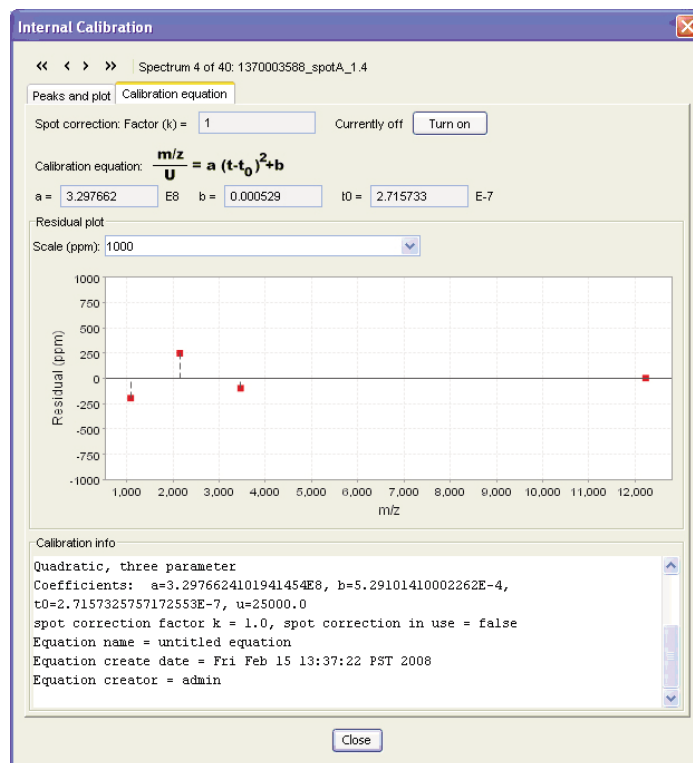
- Click the **Match Calibrants to Peak** icon . Confirm that the peaks are labeled correctly by comparing the peak names to those in the table. See Appendix A for instructions on adding missing calibrants.



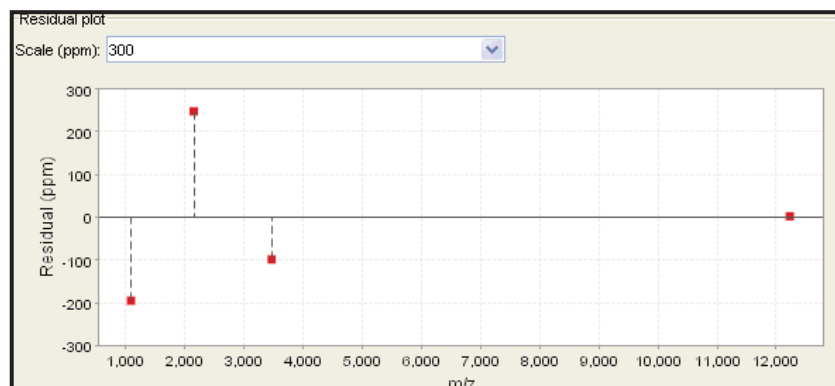
- Select **3 parameters** from the **Calibration type** drop-down list and select the **Weighted** check box. Click the **Calibrate** icon . The labels now appear in the spectra.



- Open the **Calibration equation** tab. The residual plot displays the error for each calibrant in parts per million (ppm).



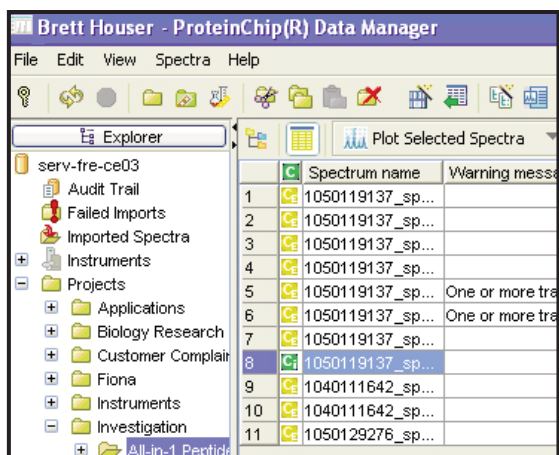
- Select **300** from the **Scale (ppm)** drop-down list. If the error of a calibrant is greater than ± 300 ppm, double-check the peak label or name to ensure proper labeling. Close the window.



- Click the **Toggle Spectrum Table** button  to view all the spectrum files within your folder. Note that the **Calibration type** icon has changed to green , indicating that the spectrum has been calibrated.

3.2 Save the Calibration Equation

- Select a calibrated spectrum (green icon) from within your **Pep Std Array** project folder.



- Save the equation by selecting **Spectra > Calibrate > Save Calibration Equation**. The **Save Calibration Equation** dialog opens.
- Enter a name for the calibration equation. Include the date of creation in the name. Click **OK**.

Note: A calibration equation can be used to externally calibrate other spectra. For best results, use the calibration equation within 1 day for mass ranges between 1 kD and 25 kD. For more information on external calibration, refer to the ProteinChip Data Manager Software Operation Manual.

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Part III: Array Preparation and Basic Data Analysis

Bar codes for the arrays used with this kit.

Array Type	Array Bar Code	Comments (Usage dates, potential errors, etc.)
ProteinChip NP20		
Premade NP20		
ProteinChip CM10		
Premade CM10		

Chapter 4

Protocol Description

SELDI and MALDI are two methods for preparing and analyzing protein and peptide samples by time-of-flight mass spectrometry (TOF-MS). In both methods, molecules are co-crystallized with UV-absorbing compounds (the matrix) before being analyzed by TOF-MS. One of the main differences between the two methods lies in the method used to prepare the sample targets:

- **MALDI uses the sandwich method.** In this method, the sample is deposited between two applications of matrix. This method is quick, but any sample contaminants such as detergents or salts must be removed prior to analysis
- **SELDI uses retentate chromatography.** In this method, protein solutions are applied directly to spots of ProteinChip arrays, which carry numerous surface chemistries that allow the selective binding of distinct protein subpopulations. Contaminants and excess proteins are then washed away and the matrix is applied. This method requires more steps than the MALDI method, but on-array washes enable the removal of contaminants and reduce the amount of abundant proteins that can suppress the detection of low abundance proteins and peptides

This manual presents two sets of protocols that allow you to perform and compare the results from both sandwich and retentate chromatography methods using the ProteinChip SELDI system:

- **Sandwich preparation of the ProteinChip NP20 array** — demonstrates the preparation and analysis of an NP20 array with human serum and *Escherichia coli* cell lysate samples using the sandwich method. The ProteinChip NP20 array has a silicon dioxide surface, which allows general capture of proteins through serine, threonine, or lysine residues. After the array is prepared, it is analyzed using the ProteinChip SELDI system, and results are compared with those obtained using the premade ProteinChip NP20 array. Use this protocol first to practice and identify any problems associated with pipetting small volumes of sample and matrix
- **Retentate chromatography preparation of the ProteinChip CM10 array** — demonstrates the preparation and analysis of a CM10 array with human serum samples. The ProteinChip CM10 array has a weak cation exchange surface, which allows selective capture of positive charges, such as lysine, arginine, and histidine residues. After the array is prepared, it is analyzed using the ProteinChip SELDI system, and results are compared with those obtained using the premade CM10 array. Use this protocol to further refine and identify any problems with sample and SELDI array preparation techniques

4.1 Thaw and Equilibrate the Samples, Reagents, and Arrays

1. Remove the following from storage and equilibrate to room temperature (5–10 min):
 - Lyophilized *E. coli* lysate
 - Human serum sample
 - ProteinChip premade NP20 and CM10 arrays¹ (1 each)
 - ProteinChip CM10 and NP20 arrays (1 each)
 - Sinapinic acid (SPA)
 - Acetonitrile
 - 1% Trifluoroacetic acid (TFA)
2. While the samples, reagents, and arrays are equilibrating, label three 15 ml tubes as follows:
 - *E. coli* (NP20)
 - Serum (NP20)
 - Serum (CM10)

4.2 Prepare and Aliquot the *Escherichia coli* Lysate (5.4 ml, 0.5 mg/ml Total Protein)

Notes: The reference samples included in this kit are supplied in excess; prepare more aliquots as needed.
Before transferring the sample, passivate the pipet tip with the sample by pipetting up and down three times.

1. Add 1.0 ml proteomics-grade water to the vial containing the lyophilized *E. coli* lysate. Pipet up and down five times to mix, and transfer to the 15 ml tube labeled *E. coli* (NP20). Repeat this step four times to transfer a total of 5 ml deionized water.
2. Add 0.4 ml deionized H₂O to the 15 ml tube, bringing the total volume to 5.4 ml.
3. Gently mix by inverting the tube several times and then vortex for 1 min.
4. Prepare five 500 µl aliquots and freeze them at –80°C for future use. Thaw each aliquot only once for use and then dispose.

¹ Do not open the packaging until you are ready to use the arrays.

4.3 Prepare and Aliquot the Serum Samples

Notes: The reference samples included in this kit are supplied in excess; prepare more aliquots as needed. Before transferring the sample, passivate the pipet tip with the sample by pipetting up and down three times.

1. Mix the serum sample by inverting the tube several times, and then briefly centrifuge to collect all the liquid at the bottom of the tube.
2. Prepare the serum dilutions for the NP20 and CM10 arrays in the labeled 15 ml tubes²:
 - Serum (NP20): 4.9 ml deionized water + 100 µl serum
 - Serum (CM10): 9.9 ml CM low-stringency buffer + 100 µl serum
3. Shake the tubes gently for 30 sec.
4. For both preparations, prepare five 500 µl aliquots and freeze them at –80°C for future use. Thaw each aliquot only once for use and then dispose using procedures appropriate for biohazardous waste.

4.4 Prepare the Matrix Solution

1. Add 200 µl acetonitrile and 200 µl 1% TFA to the tube of SPA.
2. Vortex for 5 min.
3. Centrifuge the solution at 1,000 x g for 1 min to pellet any residual solid SPA.
4. Store the vial in a dark location; the solution can be used for up to 1 week.

4.5 Download the Analysis Spreadsheet

Save the Starter Kit Analysis spreadsheet (on the CD) to your computer.

² The sample used for retentate chromatography is more dilute. It is applied using the bioprocessor, which enables application of a larger sample volume (50 times more than in the sandwich method) to compensate for this difference. Increasing the sample volume by dilution reduces variations associated with dispensing small volumes. Diluting samples also improves specific binding to the array surface.

Chapter 5

Array Preparation and Analysis Using the NP20 Array

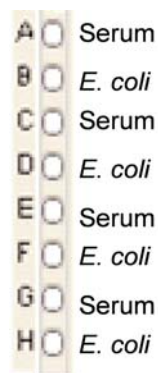
This exercise demonstrates the techniques associated with matrix and sample addition that can affect reproducibility. You prepare a ProteinChip® NP20 array with alternating spots of *E. coli* and serum for analysis using the ProteinChip reader. You then analyze the prepared array and compare your results with those obtained using the premade NP20 array.

Since this experiment has fewer array preparation steps than the CM10 (retentate chromatography) experiment, troubleshoot any “Fail” messages in this experiment before proceeding to CM10 experiments.

5.1 Prepare the ProteinChip NP20 Array

Note: During array preparation, add sample and matrix consistently to the same locations across spots A–H. Avoid touching the tip of the pipet to the surface of the array, as this can cause variations in crystal formation.

1. Remove the ProteinChip NP20 array from its packaging and record the bar code number of the array in the table on the first page of Part II of this manual. If desired, record your name and the array type on the back of the array using a laboratory marker.
2. Add 1 µl prepared matrix solution to spots A–H. Allow to dry for 5 min at room temperature.
3. Add 1 µl diluted serum (NP20) sample to spots A, C, E, and G.
4. Add 1 µl diluted *E. coli* lysate to spots B, D, F, and H.



A	Serum
B	<i>E. coli</i>
C	Serum
D	<i>E. coli</i>
E	Serum
F	<i>E. coli</i>
G	Serum
H	<i>E. coli</i>

5. Add 1 µl prepared matrix solution to all spots. Change pipet tips between applications to avoid sample carry-over.
6. Allow the array to dry for 20 min at room temperature.

5.2 Load the Array and Gain Control of the Reader

5.2.1 Enterprise Edition

The Enterprise Edition reader uses a cassette to load up to 12 arrays at a time.

1. Place the NP20 array you prepared into slot #1 of the cassette.
2. Remove the premade NP20 array from its packaging and insert it into slot #2, with spot H entering the cassette first. Record the bar code of the array in the table on the cover of this manual.
3. Fill the empty slots of the cassette with blank arrays.
4. Place the cassette into the reader and close the lid. The reader moves the cassette into position while reading the bar codes of the arrays. The arrays in the cassette are now in position to be loaded into the reader. Each array automatically loads one at a time when a protocol is run.
5. In the **Explorer** pane of ProteinChip data manager software, open the **Instruments** folder and select the reader from the list. The **Home** tab opens.

6. In the list of array bar codes, confirm the NP20 array is in location #1 and the premade NP20 array is in location #2.
7. Click the **Start** button  to gain control of the reader. The **Manual Mode** tab opens, indicating the arrays are in position and ready to be used.

5.2.2 Personal Edition

The Personal Edition reader analyzes one array at a time. Use the following instructions for loading each array. Load your prepared NP20 array first before loading the premade NP20 array.

1. In the **Explorer** pane of ProteinChip data manager software, click the Instruments icon and select the reader from the list. The **Home** tab opens.
2. Click the **Start** button  to gain control of the reader.
3. Click the **Set Array Barcode** button . The **Set Array Barcode** dialog opens. Enter the bar code on the array and click **Set Barcode**.
4. Click the **Door Unlock** icon  in the toolbar to release the vacuum.
5. Once the blue light on the reader begins to flash, open the door and slide the array into the reader, with spot H entering first.
6. Check the gasket for any lint or dust that may prevent the reader from obtaining the necessary vacuum. If necessary, use a methanol- or ethanol-dampened, lint-free cloth to wipe the gasket clean.
7. Close the door and click the **Door Lock** icon . The reader prepares the array for analysis. The **Set Array Barcode** dialog opens and displays the current bar code. Click **Set Barcode**. The reader prepares the array for analysis.

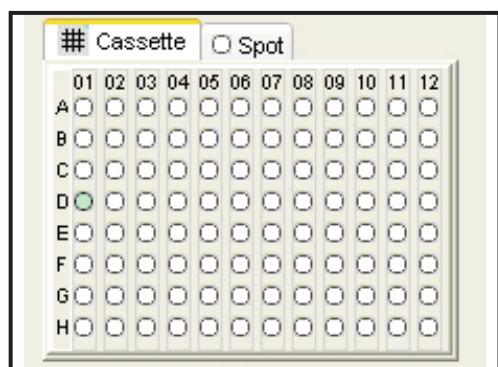
5.3 Optimize the Laser Energy Setting (7.2 kD Peak, 50–100 μ A)

Data are collected from arrays using preset data collection protocols. For the best data quality, always optimize the parameters and settings of the data collection protocols for each sample type. Optimize the laser energy settings for different sample types and to account for aging of reader components. Optimize the laser energy settings of a protocol each day, even when using the same sample type.

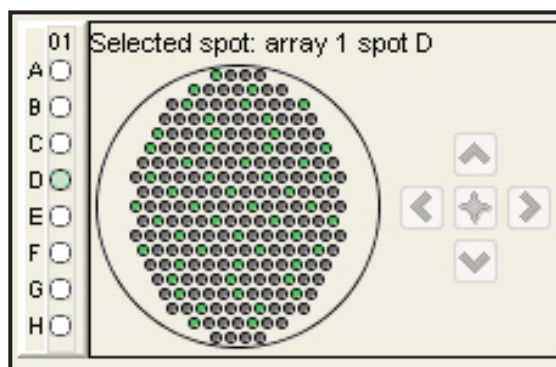
You imported the protocol required for analyzing the NP20 and CM10 arrays (Starter Kit.pdx) in Part I. Here, you optimize the laser energy setting such that the intensity for the peak at 7.2 kD is 50–100 μ A. An example of this procedure is provided in the instructional video.

1. Open the **Protocol Mode** tab. In the **Acquisition Protocol** section, select the **Starter Kit** protocol from the **Name** drop-down list.
2. Create a subfolder for the NP20 data by clicking again on the **Folder** button . The **Select Folder** dialog opens. Double-click on the **Projects** folder and then right-click on **Starter Kit Data**. Create a new project subfolder, **My NP20 Array**.
3. In Enterprise Edition software, the cassette map shows the distribution of arrays in the cassette. Columns 1–12 correspond to arrays 1–12. The NP20 array is represented by column 1. Under column 1 (both software versions), select spot **D**. The spot turns green.

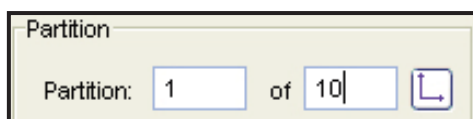
Enterprise Edition



Personal Edition



- Set the partition to **1 of 10** and click the **Start Running Protocol** icon . The icon turns red, and the reader loads the array.



- Once the spot is analyzed, the spectrum appears in the **Spectrum** section of the **Protocol Mode** tab (about 1 min).
- Zoom in to the peak at 7.2 kD by clicking and dragging on either side of the peak, and then determine the intensity in μA .
- Adjust the laser energy so that the peak is 50–100 μA .

To change the laser setting, use the **Protocol Setting** tab. Adjust the laser energy using the **Data Shots(s) Energy** box. Adjust the warming shots using the **Warming Shots(s) Energy** box. Leave the **Add** box unchecked.

- If the intensity is $<50 \mu\text{A}$, increase the laser energy setting by 10% and increase the warming shot(s) energy to 10% higher than the data shot(s) energy
 - If the intensity is $>100 \mu\text{A}$, decrease the laser energy setting by 10% and decrease the warming shot(s) energy to 10% higher than the data shot(s) energy
- Run the protocol again using partition **2 of 10**. Changing the partitions allows the reader to collect data from a fresh location.
 - Ensure spot **D** is selected and click the **Start Running Protocol** icon to run the protocol with the new settings.
 - Repeat steps 7–10, using a new partition each time (for example, the next run would be 3 of 10) until you obtain a peak intensity of 50–100 μA .
 - Apply the optimized protocol to all spots on the array.

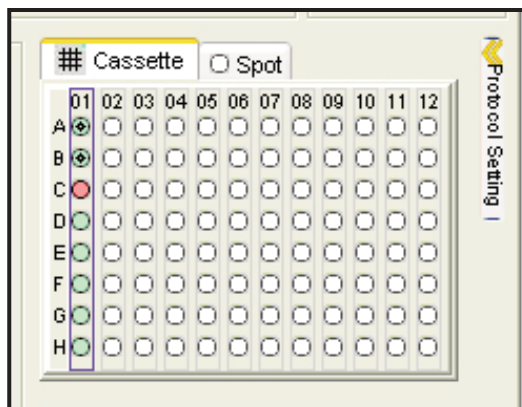
5.4 Collect Data From the NP20 Array

Using the new data collection settings, analyze your entire NP20 array.

- Change the partitions back to **1 of 4**.
- In the cassette map, select all 8 spots on the array by clicking on each spot. The spots turn green. Click the **Start Running Protocol** icon .

3. The reader analyzes each spot (2 min each). The following icons indicate the status of each spot:

-  Data have been collected
-  Data are being collected
-  Data are going to be collected

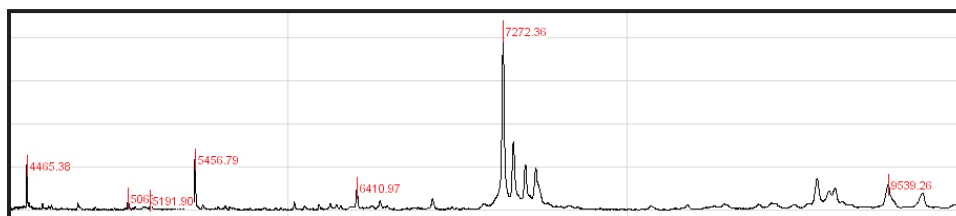


The status of each spot is also shown below the spectrum, along with information about the partition and protocol used.

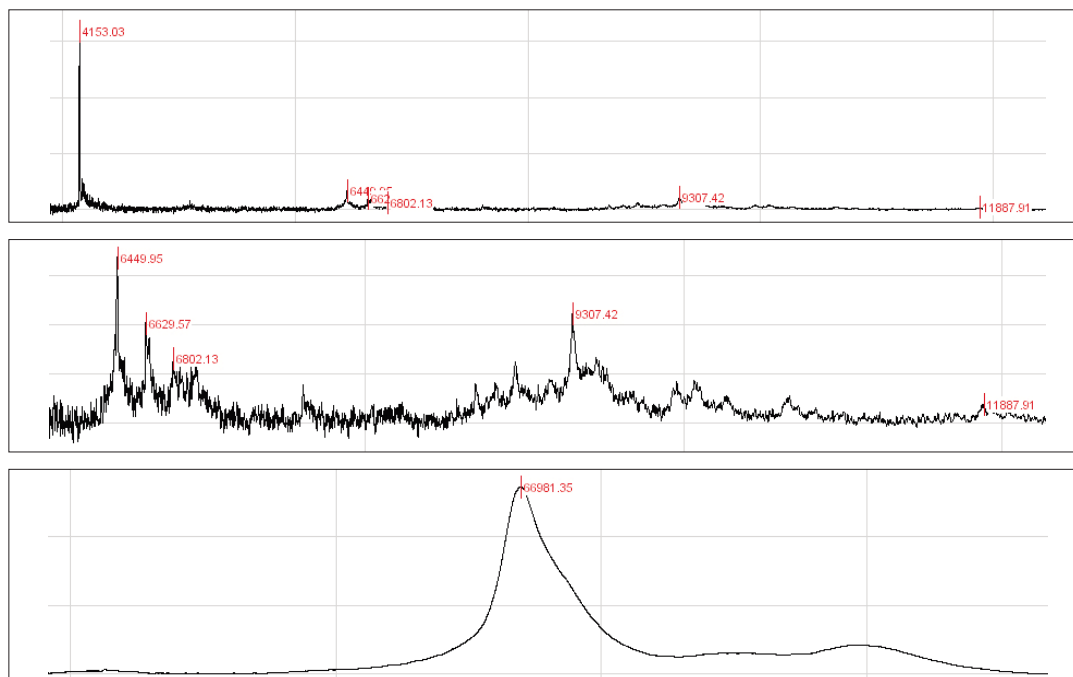
		Barcode	Array	Spot	Partition	Protocol	Spectrum Name	Completion Time
1	✓	1020000011	1	A	1 / 4	Starter Kit	1020000011_spotA_1_4	Fri May 09 13:52:10 PDT 2008
2	✓	1020000011	1	B	1 / 4	Starter Kit	1020000011_spotB_1_4	Fri May 09 13:53:09 PDT 2008
3		1020000011	1	C	1 / 4	Starter Kit		
4		1020000011	1	D	1 / 4	Starter Kit		
5		1020000011	1	E	1 / 4	Starter Kit		
6		1020000011	1	F	1 / 4	Starter Kit		
7		1020000011	1	G	1 / 4	Starter Kit		
8		1020000011	1	H	1 / 4	Starter Kit		

The spectra obtained from both sample types should resemble those below.

E. coli



Serum



5.5 Collect Data From the Premade NP20 Array

Analyze the premade NP20 array supplied with the kit.

1. Load the array:
 - If using the Enterprise Edition, the array should be in location #2 of the cassette
 - If using the Personal Edition, eject your NP20 array from the reader by clicking the **Door Unlock** icon  in the toolbar to release the vacuum. Load the premade NP20 array into the reader according to the instructions in Section 5.2.2
2. Under the **Protocol Mode** tab, select the **Starter Kit** protocol from the **Name** drop-down list in the **Acquisition Protocol** section.
3. Create a subfolder for the data by clicking on the **Folder** button . The **Select Folder** dialog opens. Double-click on the **Projects** folder and then right-click on **Starter Kit Data**. Create a new project subfolder, **Premade NP20 Array**.
4. Using spot **D**, optimize the laser energy settings as described in Section 5.3.
5. Collect data from the premade NP20 array as described in Section 5.4.
6. Once analysis is complete, eject the cassette or array from the reader by selecting **Instrument > Eject Cassette** (Enterprise Edition) or by clicking the **Door Unlock** icon  in the toolbar (Personal Edition).

Chapter 6

NP20 Array Data Analysis

This chapter summarizes the basic steps to follow to analyze your data:

- Label a set of peaks on each array, export these data to the analysis spreadsheet, and analyze the reproducibility of their intensities across the spots and arrays you analyzed
- Perform automated peak counts of both low-mass and high-mass peptides and export these data to the analysis spreadsheet to calculate the reproducibility of your analysis
- Gauge the performance of the arrays you prepared against the premade arrays

These instructions assume that you know how to select and plot spectra and how to label peaks. For more information about these functions, refer to Part I of the ProteinChip® Starter Kit Instruction Manual.

For a summary of the data analysis parameters and settings used, refer to Appendix B.

6.1 Compare Peak Intensity Results

Label a set of peaks on each array, export these data to the analysis spreadsheet (according to array type), and analyze the reproducibility of their intensities across the spots and arrays you analyzed.

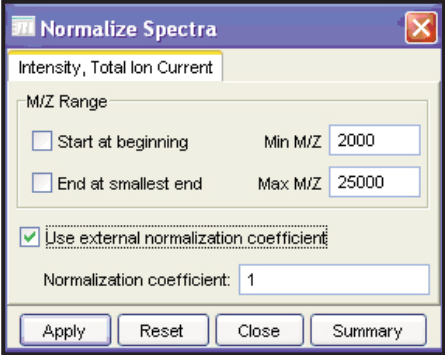
1. In the **Explorer** pane, locate and open the **My NP20 Array** folder. A list of spectra appears.

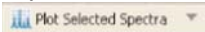
	C	Spectrum name	Date collected	...	...	.	.	Spot name	Frac
1		1170308422_spotD_1.10	2008-01-28 18:...	1				D	
2		1170308422_spotD_2.10	2008-01-28 18:...	1				D	
3		1170308422_spotA_1.4	2008-01-28 18:...	1				A	
4		1170308422_spotB_1.4	2008-01-28 18:...	1				B	
5		1170308422_spotC_1.4	2008-01-28 18:...	1				C	
6		1170308422_spotD_1.4	2008-01-28 18:...	1				D	
7		1170308422_spotE_1.4	2008-01-28 18:...	1				E	
8		1170308422_spotF_1.4	2008-01-28 18:...	1				F	
9		1170308422_spotG_1.4	2008-01-28 18:...	1				G	
10		1170308422_spoth_1.4	2008-01-28 18:...	1				H	

2. Delete the spectra used for laser optimization. To find these spectra, sort the spectra by **Date collected**. Spectra collected during optimization appear for spot D and with a **Date collected** value that precedes that for spot A. They may also have a spectrum name ending in "1.10" (1 of 10 partitions). Select and delete these spectra using the Delete key on your computer keyboard.

6.1.1 Label Peaks

1. Select the remaining 8 spectra and click the **Normalize Spectra** icon . The **Normalize Spectra** dialog opens.
2. Enter the following values and parameters and click **Apply**.



3. Label the 7 peaks in the table below in both sets of spectra. First, select the serum spectra (hold the Ctrl key and click on spots A, C, E, and G) and plot them by selecting **Plot Selected Spectra** from the **Plot Spectra** drop-down . Label the peaks indicated below (refer to the video tutorial for information on how to label peaks). Repeat these steps with the *E. coli* spectra (spots B, D, F, and H). Refer to the video tutorial or to Part I of the instruction manual for information on how to label peaks.

Peaks selected for labeling.

Serum (kD)	<i>E. coli</i> (kD)
4.15	4.46
6.44	5.06
6.62	5.19
6.80	5.45
9.30	6.41
11.88	7.27
66.98	9.53

6.1.2 Export Peaks

1. Export the data. In the **Explorer** pane, open the **Projects > Starter Kit Data > My NP20 Array** folder and select the 8 spectra for export.
2. Click the **Export Spectra** button  in the toolbar. The **Export Spectra** dialog appears.
3. Select **Peak Information** and click **Export**. The **Peak Information Export** dialog appears.

4. Select the following parameters for export and click **OK**:

Spectrum Fields

- Spectrum #
- Spectrum name

Select Peak Fields

- Peak #
- Substance Mass
- Intensity

5. Name the .csv file ***My NP20 Array***.
6. Select the data from the .csv file and paste the values into the Starter Kit Analysis spreadsheet under the **NP20 Array Data** tab. Paste the values for your NP20 array into the columns designated for **My NP20 Array: Manually Selected Peaks**.
7. Repeat all steps starting from 6.1 with the spectra in the **Premade NP20 Array** folder. Name the .csv file ***Premade NP20 Array*** and paste the values into the columns designated for **Premade NP20 Array: Manually Selected Peaks**.
8. The spreadsheet calculates the reproducibility (coefficients of variation, CVs) for the intensities of the chosen peaks. View the results under the **NP20 Results** tab.

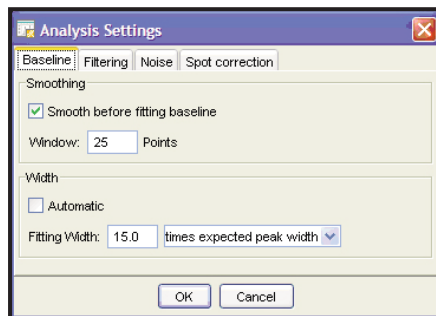
6.2 Compare Peak Count Results

Perform automated peak counts on all spectra to further evaluate sample and matrix addition. First, perform an automated peak count on the spectra in the low-mass region (2–25 kD). Then, perform an automated peak count in the high-mass region (25–200 kD)

Note: Perform peak counts on each array separately, and export the data according to array type.

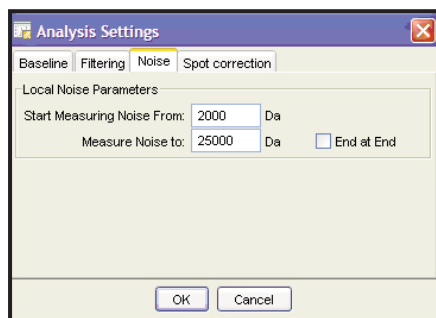
6.2.1 Low-Mass Region (2–25 kD)

1. Select the 8 spectra in the **My NP20 Array** folder and click the **Analysis Settings** icon . The **Analysis Settings** dialog opens, displaying the baseline correction parameters. Enter the values and select the parameters shown below. Do not click **OK**.



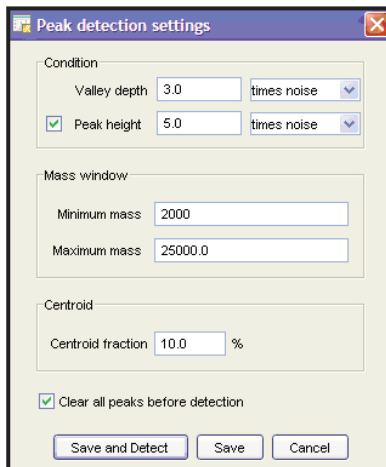
The **Analysis Settings** dialog box is shown with the **Baseline** tab selected. The **Smoothing** section has ☒ **Smooth before fitting baseline** and **Window:** 25 **Points**. The **Width** section has ☐ **Automatic** and **Fitting Width:** 15.0 **times expected peak width**. **OK** and **Cancel** buttons are at the bottom.

2. In the **Analysis Settings** dialog, open the **Noise** tab. Deselect **End at End** and enter the values shown below. Click **OK**.



The **Analysis Settings** dialog box is shown with the **Noise** tab selected. The **Local Noise Parameters** section has **Start Measuring Noise From:** 2000 **Da**, **Measure Noise to:** 25000 **Da**, and ☐ **End at End**. **OK** and **Cancel** buttons are at the bottom.

3. Select all 8 spectra in the **My NP20 Array** folder and plot them by selecting **Plot Selected Spectra** from the **Plot Spectra** drop-down . Click the **Detect Peaks** icon . The **Peak detection settings** dialog opens. Enter the values and select the parameters shown below, then click **Save and Detect**.

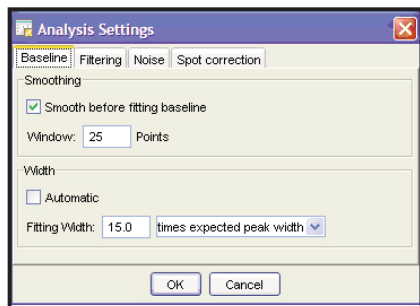


The **Peak detection settings** dialog box is shown. The **Condition** section has **Valley depth:** 3.0 **times noise** and ☒ **Peak height:** 5.0 **times noise**. The **Mass window** section has **Minimum mass:** 2000 and **Maximum mass:** 25000.0. The **Centroid** section has **Centroid fraction:** 10.0 **%**. ☒ **Clear all peaks before detection** is checked. **Save and Detect**, **Save**, and **Cancel** buttons are at the bottom.

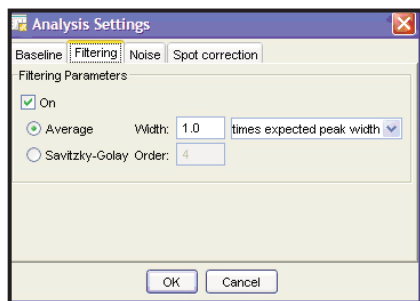
4. Repeat steps 1–3 for the 8 spectra from the **Premade NP20 Array** folder.

6.2.2 High-Mass Region (25–200 kD)

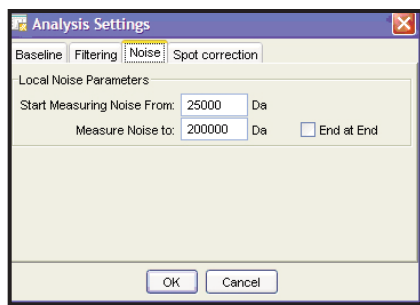
1. Select the 8 spectra and click the **Analysis Settings** icon . The **Analysis Settings** dialog opens, displaying the baseline correction parameters. Enter the values and select the parameters shown below. Do not click **OK**.



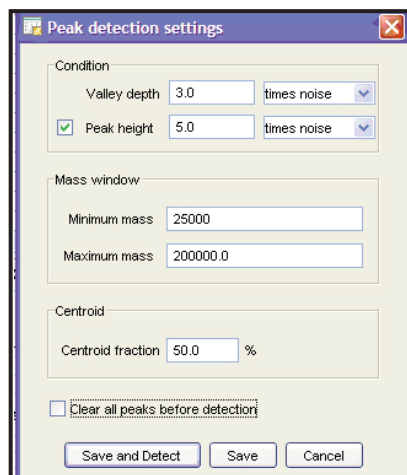
2. In the **Analysis Settings** dialog, open the **Filtering** tab and enter the values and select the parameters as below. Do not click **OK**.



3. Open the **Noise** tab. Deselect **End at End** and enter the values and select the parameters as below. Click **OK**.



4. Select and plot all 8 spectra again and click the **Detect Peaks** icon . The **Peak detection settings** dialog opens. Enter the values and select the parameters as below (remember to deselect **Clear all peaks before detection**), then click **Save and Detect**.



5. Repeat all steps starting from 6.2 for the spectra in the **Premade NP20 Array** folder.

6.2.3 Export the Data

Export the data from each array type separately.

1. Select all the spectra in the **My NP20 Array** folder to be exported and click the **Export Spectra** button  in the toolbar. The **Export Spectra** dialog appears.
2. Select **Peak Information** and click **Export**. The **Peak Information Export** dialog appears.
3. Select the following parameters for export and click **OK**:

Spectrum Fields

- Spectrum #
- Spectrum name

Select Peak Fields

- Peak #
- Substance Mass
- Intensity

4. Name the .csv file **My NP20 Array PC**.
5. Select the data from the .csv file and paste the values into the Starter Kit Analysis spreadsheet under the **NP20 Array Data** tab. Paste the values for your NP20 array into the columns designated for **My NP20 Array: Automatic Peak Detection**.
6. Repeat steps 1–5 with the spectra in the **Premade NP20 array** folder. Name the .csv file **Premade NP20 Array PC** and save the values to the columns designated for **Premade NP20 Array: Automatic Peak Detection**.
7. The spreadsheet calculates the reproducibility (CVs) for the peak counts. View the results under the **NP20 Results** tab.

6.3 Expected Results

Measurement	Expected Value		Assessment of...
	<i>E. coli</i>	Serum	
Peak intensity (pooled CV)	<30%	<35%	Checks reproducibility of intensities of selected peaks across all spectra of sample type
Peak count (# peaks)	>45	>10	Checks number of peaks detected; compare to number expected

If your values differ from those in the table, inspect the spectra for proper peak labeling and any distinct anomalies in their appearance (for example, missing or unexpected peaks). Refer to Appendix C, Troubleshooting for more information.

Chapter 7

SELDI Preparation and Analysis Using CM10 Arrays

This protocol demonstrates the preparation and analysis of a ProteinChip® CM10 array (weak cation exchange) with human serum samples. After preparing the array, you analyze it using the ProteinChip SELDI system and compare your results with those obtained using a premade CM10 array, which already contains serum samples and matrix. Use this protocol to practice and identify any potential problems with sample and array preparation technique.

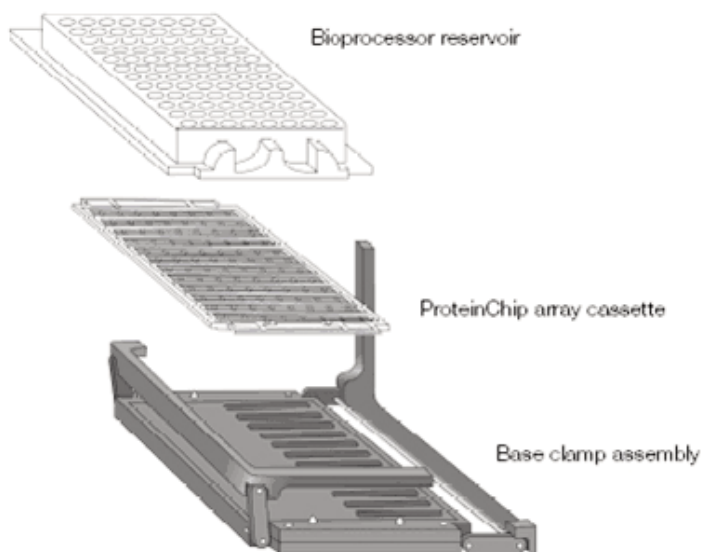
You will use a diluted human serum sample and the ProteinChip cassette-compatible bioprocessor for array preparation. The ProteinChip bioprocessor allows application of larger sample volumes to a ProteinChip array surface. Increasing the sample volume by dilution reduces variations associated with dispensing small volumes and improves specific binding to the array surface. The ProteinChip cassette-compatible bioprocessor enables simultaneous processing of up to 12 ProteinChip arrays and is compatible with robotic or liquid-handling systems used for 96-well format assays.

During array preparation, add sample and matrix consistently to the same locations across spots A–H. Avoid touching the tip of the pipet to the surface of the array, as this can cause variation in crystal formation.

7.1 Prepare the ProteinChip CM10 Array

7.1.1 Equilibrate the Array

1. Remove the ProteinChip CM10 array from its packaging. Record the bar code number of the array in the table the first page of Part II of this manual. If desired, record your name and the array type on the back of the array using a laboratory marker.
2. Handling the array by its edges, gently push it into slot #1 of the cassette, with spot H entering the cassette first. Insert a blank or used array into slot #2.
3. Place the cassette into a ProteinChip cassette-compatible bioprocessor. Assemble the processor with the reservoir.



4. Equilibrate the array by adding 150 μ l CM low-stringency buffer to each spot. Ensure there are no bubbles at the bottoms of the bioprocessor wells. To remove bubbles, use a clean pipet tip to dislodge them (do not touch the array surface) or centrifuge the bioprocessor for 1 min at 250 x g.
5. Shake the bioprocessor on a shaker for 5 min at room temperature (for example, on a MicroMix shaker at a setting of 20/5), then remove the buffer by firmly tapping the bioprocessor upside down over a paper towel several times.
6. Repeat steps 4 and 5.

7.1.2 Apply and Bind Sample to the Array

1. Pipet 50 μ l of the serum (CM10) preparation into each well of the CM10 array. Avoid contact between the pipet tip and the array surface.
2. Inspect the wells for bubbles, and remove bubbles if necessary.
3. Cover the bioprocessor with sealing film and incubate on a shaker (for example, on a MicroMix shaker at a setting of 20/5) for 30 min at room temperature.

Note: The details of this dilution and binding step may require optimization for different sample types.

7.1.3 Wash the Array

1. Remove the sample from the wells by firmly tapping the bioprocessor upside down over a paper towel several times. Dispose of the paper towel appropriately.
2. Add 150 μ l CM10 low-stringency buffer to the wells and shake for 5 min at room temperature.
3. Remove the buffer by holding the bioprocessor upside down over a paper towel and tapping on it firmly several times.
4. Repeat steps 2 and 3 two times.

5. Add 200 μ l deionized water to the wells and remove it immediately by holding the bioprocessor upside down over a paper towel and tapping on it firmly several times. Repeat.

7.1.4 Apply the Matrix

1. Remove the CM10 array from the bioprocessor assembly:
 - a. Place the bioprocessor on a lab bench or other flat surface with the solid (back) surface of the bioprocessor down.
 - b. Unlatch the hinged leg assemblies and remove the reservoir from the base clamp assembly. Hold the base clamp assembly down with one hand, while lifting the reservoir from the base with the other hand.
2. Remove any excess water from the arrays by blotting with a lint-free lab wipe.
3. Allow the arrays to air-dry for 10–20 min. Keep the cassette flat in the bioprocessor by using the cassette hold-down frame provided with this assembly.
4. Add 1.0 μ l of the SPA matrix to each spot. Allow to dry at room temperature for at least 10 min. Repeat.

7.2 Load the Array and Gain Control of the Reader

These steps are similar to those used with the NP20 arrays in Chapter 5. Refer to Section 5.2 for more details, if necessary.

1. Replace the blank array in slot #2 of the cassette with the premade CM10 array, with spot H entering the cassette first. Record the bar code of the premade array in the table on the cover of this manual.
2. Fill the empty slots of the cassette with blank arrays.
3. Place the cassette into the reader and close the lid.
4. In the **Explorer** pane of ProteinChip data manager software, open the **Instruments** folder and select the reader from the list. The **Home** tab opens.
5. In the list of array bar codes, confirm the CM10 array is in location #1 and the premade CM10 array is in location #2.
6. Click the **Start** button  to gain control of the reader. The **Manual Mode** tab opens, indicating the arrays are in position and ready to be used.

Follow the instructions in Section 5.2.2 to load and analyze your CM10 array using the Personal Edition reader.

7.3 Optimize the Laser Energy Setting (8.1 kD Peak, 50–200 μ A)

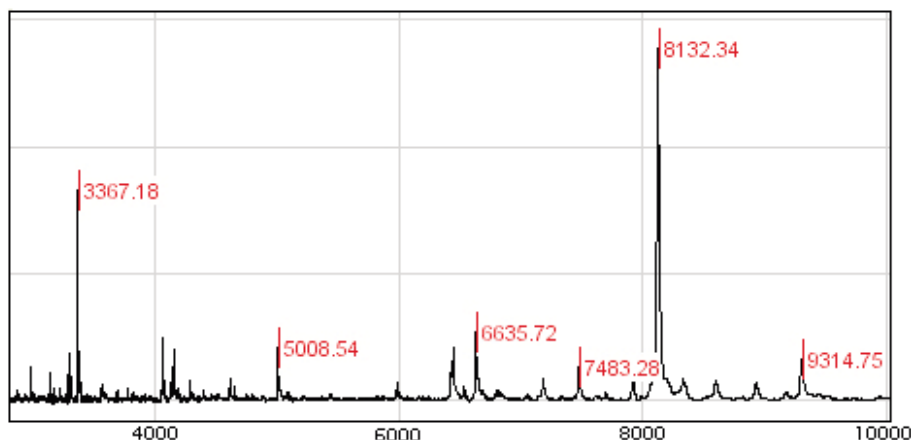
Follow the instructions detailed in Section 5.3 to optimize the laser energy such that the intensity for the dominant peak at 8.1 kD is 50–200 μ A. Create and save your data to a new project subfolder **Starter Kit Data > My CM10 Array**.

7.4 Collect Data From the CM10 Array

Using the new data collection settings, analyze your entire CM10 array.

1. Change the partitions back to **1 of 4**.
2. In the cassette map, select all 8 spots of the array by clicking on each spot. The spots turn green. Click the **Start Running Protocol** icon . The reader analyzes each spot (2 min each).

The 3,000–10,000 m/z region should resemble that below.



7.5 Collect Data From the Premade CM10 Array

Analyze the ProteinChip premade CM10 array supplied with the kit.

- Load the array:
 - If using the Enterprise Edition, the array should be in location #2 of the cassette
 - If using the Personal Edition, eject your CM10 array from the reader by clicking the **Door Unlock** icon  in the toolbar to release the vacuum. Load the premade CM10 array into the reader, following the instructions in Section 5.2.2. Record the bar code of the premade CM10 array in the table on the first page of Part II of this manual
- In the **Acquisition Protocol** section of the **Protocol Mode** tab, select the **Starter Kit** protocol from the **Name** drop-down list.
- Create a subfolder for the premade CM10 data by clicking on the **Folder** button . The **Select Folder** dialog opens. Double-click on the **Projects** folder and then right-click on **Starter Kit Data**. Create a new project subfolder, **Premade CM10 Array**.
- Using spot D, optimize the laser energy settings as described in Section 7.3.
- Collect data from the premade CM10 array as described in Section 7.4.
- Once analysis is complete, eject the cassette or array from the reader by selecting **Instrument > Eject Cassette** (Enterprise Edition) or by clicking the **Door Unlock** icon  in the toolbar (Personal Edition).

Chapter 8

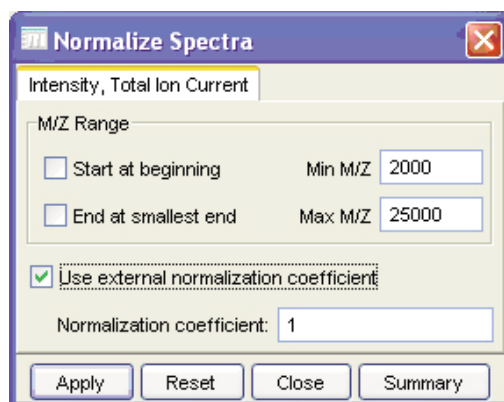
CM10 Array Data Analysis

This chapter summarizes the basic steps to follow to analyze your data. For more details on these steps, please refer to Chapter 4. For a summary of the data analysis parameters and settings used, refer to Appendix B.

8.1 Compare Peak Intensity Results

Perform data processing and analysis steps and export the results according to array type.

- In the **Explorer** pane, locate and open the **My CM10 Array** folder. A list of spectra appears.
- Delete the spectra used for laser optimization.
- Select the remaining 8 spectra and click the **Normalize Spectra** icon . The **Normalize Spectra** dialog opens.
- Enter the values and select the parameters below and click **Apply**.

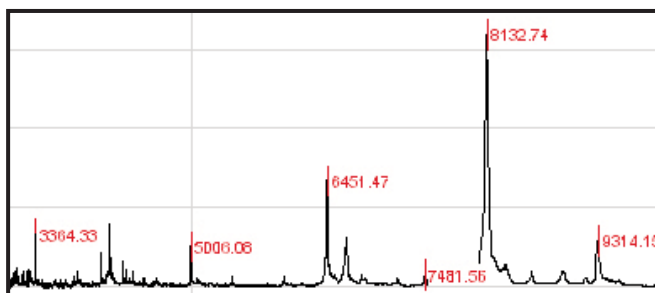


5. Select the 8 spectra again and plot them by selecting **Plot Selected Spectra** from the **Plot Spectra** drop-down .

8.1.1 Label Peaks

1. Label the following peaks in the spectra (refer to the video tutorial or to Part I of the manual for information on how to label peaks):

Serum (kD)
3.36
5.00
6.45
7.48
8.13
9.31
66.47



8.1.2 Export Peaks

1. Export the data. In the **Explorer** pane, open the **Projects > Starter Kit Data > My CM10 Array** folder and select the spectra to be exported. Click the **Export Spectra** button  in the toolbar. The **Export Spectra** dialog appears.
2. Select **Peak Information** and click **Export**. The **Peak Information Export** dialog appears.
3. Select the following parameters for export and click **OK**:

Spectrum Fields

- Spectrum #
- Spectrum name

Select Peak Fields

- Peak #
- Substance Mass
- Intensity

4. Name the .csv file **My CM10 Array**.
5. Select the data from the .csv file and paste the values into the **Starter Kit Analysis spreadsheet** under the **CM10 Array Data** tab. Paste the values for your CM10 array into the columns designated for **My CM10 Array: Manually Selected Peaks**.

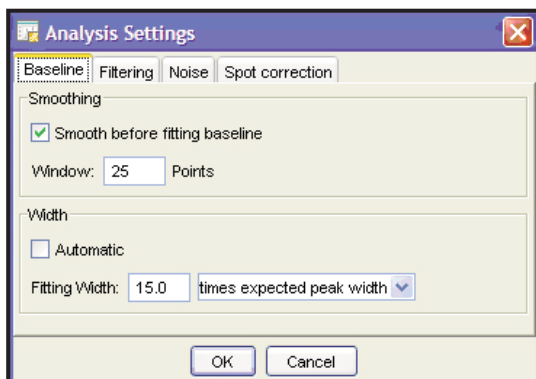
- Repeat steps starting with 8.1 with the spectra in the **Premade CM10 Array** folder. Name the .csv file **Premade CM10 Array** and paste the values into the columns designated for **Premade CM10 Array: Manually Selected Peaks**.
- The spreadsheet calculates the reproducibility (coefficients of variation, CVs) for the intensities of the chosen peaks. View the results under the **CM10 Results** tab.

8.2 Compare Peak Counts

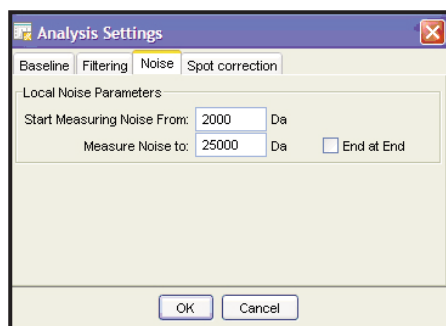
Perform automated peak counts on all spectra to further evaluate sample and matrix addition. First, perform an automated peak count on the spectra in the low-mass region (2–25 kD). Then, perform an automated peak count in the high-mass region (25–250 kD).

8.2.1 Low-Mass Region (2–25 kD)

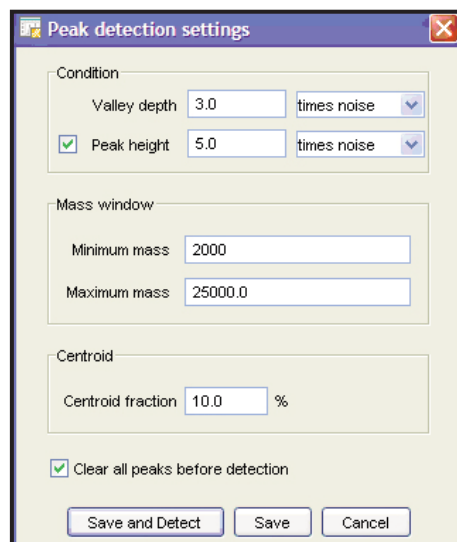
- Select the 8 spectra in the **My CM10 Array** folder and click the **Analysis Settings** icon . The **Analysis Settings** dialog opens. Enter the values and select the parameters shown below. Do not click **OK**.



- In the **Analysis Settings** dialog, open the **Noise** tab. Perform noise filtering by deselecting **End at End** and entering the values shown below. Click **OK**.



3. Select and plot the 8 spectra and then click the **Detect Peaks** icon . The **Peak detection settings** dialog opens. Enter the values and select the parameters shown below, then click **Save and Detect**.



Peak detection settings

Condition

Valley depth: 3.0 times noise

☒ Peak height: 5.0 times noise

Mass window

Minimum mass: 2000

Maximum mass: 25000.0

Centroid

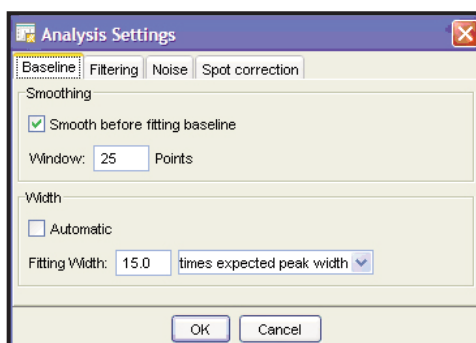
Centroid fraction: 10.0 %

☒ Clear all peaks before detection

Save and Detect Save Cancel

8.2.2 High-Mass Region (25–200 kD)

1. Select and plot all 8 spectra and click the **Analysis Settings** icon . The **Analysis Settings** dialog opens, displaying the baseline correction parameters. Enter the values and select the parameters shown below. Do not click **OK**.



Analysis Settings

Baseline Filtering Noise Spot correction

Smoothing

☒ Smooth before fitting baseline

Window: 25 Points

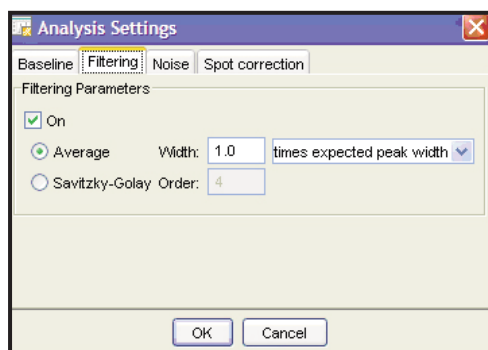
Width

☐ Automatic

Fitting Width: 15.0 times expected peak width

OK Cancel

2. In the **Analysis Settings** dialog, open the **Filtering** tab and enter the values and select the parameters as below. Do not click **OK**.



Analysis Settings

Baseline Filtering Noise Spot correction

Filtering Parameters

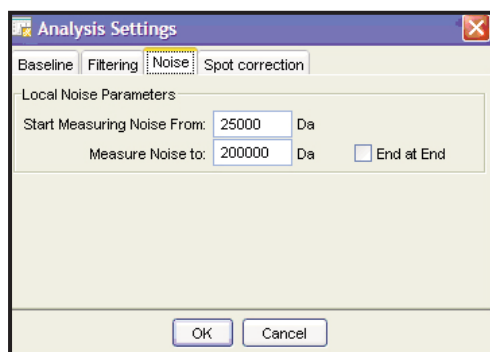
☒ On

☒ Average Width: 1.0 times expected peak width

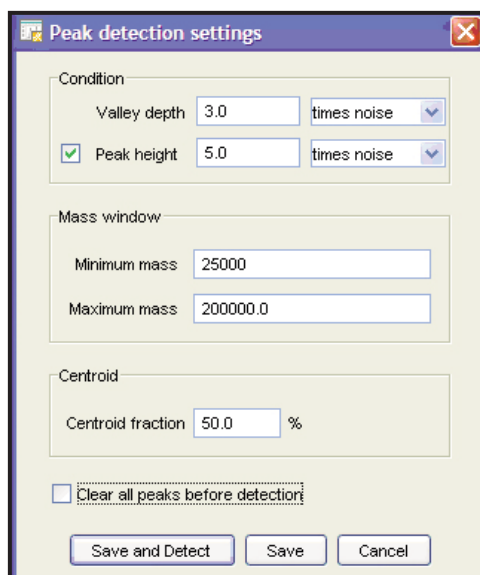
☐ Savitzky-Golay Order: 4

OK Cancel

- Open the **Noise** tab. Deselect **End at End** and enter the values and select the parameters as below. Click **OK**.



- Select and plot the 8 spectra again and click the **Detect Peaks** icon . The **Peak detection settings** dialog opens. Enter the values and select the parameters as below (remember to deselect **Clear all peaks before detection**), then click **Save and Detect**.



8.2.3 Export the Data

- Select all the spectra to be exported and click the **Export Spectra** button  in the toolbar. The **Export Spectra** dialog appears.

Select **Peak Information** and click **Export**. The **Peak Information Export** dialog appears.

Select the following parameters for export and click **OK**:

Spectrum Fields

- Spectrum #
- Spectrum name

Select Peak Fields

- Peak #
- Substance Mass
- Intensity

- Name the .csv file **My CM10 Array PC**.

3. Select the data from the .csv file and paste the values into the Starter Kit Analysis spreadsheet under the **CM10 Array Data** tab. Paste the values for your CM10 array into the columns designated for **My CM10 Array: Automatic Peak Detection**.
4. Repeat steps starting with Section 8.2 with the spectra in the **Premade CM10 Array** folder. Name the .csv file **Premade CM10 Array** PC and paste the values into the columns designated for **Premade CM10 Array: Automatic Peak Detection**.
5. The spreadsheet calculates the average peak counts. View the results under the **Results** tab.

8.3 Expected Results

Measurement	Expected Value	Assessment of...
Peak intensity (pooled CV)	<25%	Reproducibility of intensities of selected peaks across all spectra
Peak count (# peaks)	>35	Number of peaks detected; compared to number expected

If your values differ from those in the table, inspect the spectra for proper peak labeling and any distinct anomalies in their appearance (for example, missing or unexpected peaks). Refer to Appendix C, Troubleshooting for more information.

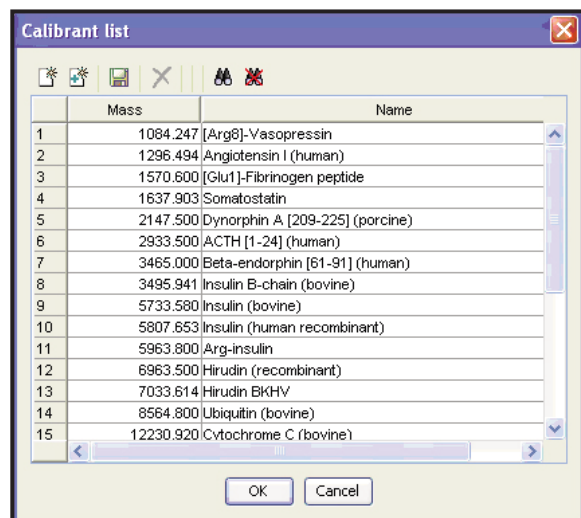
Appendix A

Adding Mass Calibrants

ProteinChip® data manager software includes a set of standard calibrant masses.

To add new calibrants:

1. Click the **Calibrants...** button  in the calibration toolbar. The **Calibrant list** dialog appears.



2. Click the **New Row** icon  to add a new calibrant to the list, or the **New Multiple Rows**  icon to add a series of calibrants.
3. Click the **Delete Rows** icon  (available when a calibrant is selected) to remove calibrants.

Appendix B

Analysis Summary Sheet

Data Collection Folders

Projects

- > Starter kit Data
 - > My NP20 Array
 - > Premade NP20 Array
 - > My CM10 Array
 - > Premade CM10 Array

Data Analysis Settings

Measurement	All-Mass Range (2–200 kD)	Low-Mass Range (2–25 kD)	High-Mass Range (25–200 kD)
Peak Intensity (Pooled CV)			
Normalization		External normalization coefficient = 1	
MALDI <i>E. coli</i> peaks	4.46, 5.06, 5.19, 5.45, 6.41, 7.27, 9.53 kD		
MALDI serum peaks	4.15, 6.44, 6.62, 6.80, 9.30, 11.88, 66.98 kD		
SELDI serum peaks	3.36, 5.00, 6.45, 7.48, 8.13, 9.31, 66.47 kD		
Peak Counts			
Analysis settings			
Baseline	15 x expected peak width (EPW)	15 x expected peak width (EPW)	15 x expected peak width (EPW)
Filtering		Default	1 x EPW
Noise		2,000 to 25,000 Da (Deselect End at End)	25,000 to 200,000 Da (Deselect End at End)
Peak detection settings			
Valley depth	3	3	3
Peak height	5	5	5
Mass window		2,000 to 25,000 Da	25,000 to 200,000 Da
Centroid		10%	50%
Clear all peaks before detection		Yes	No

Appendix C

Troubleshooting

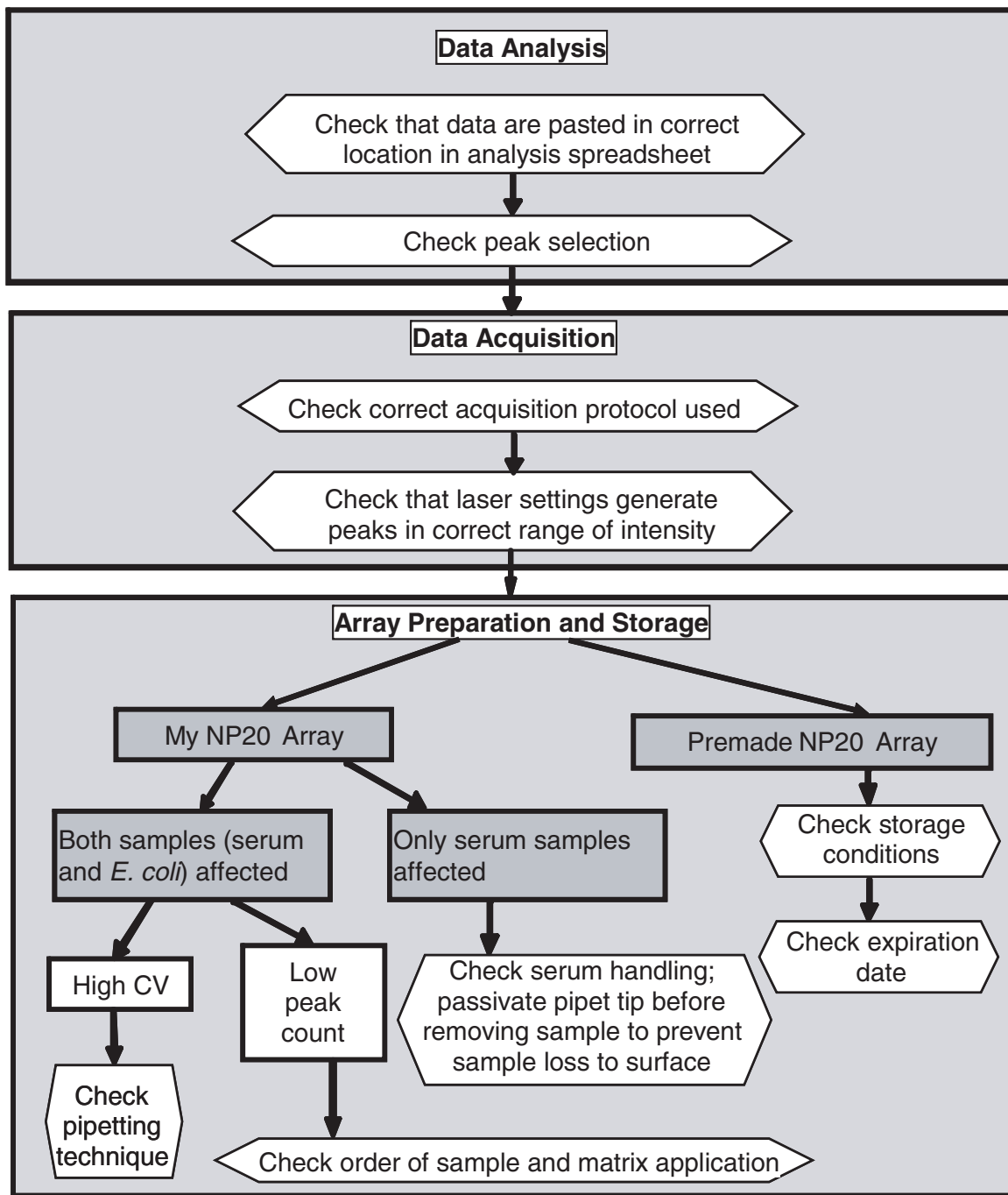
A “Fail” message appears in the analysis spreadsheet whenever CVs are greater or peak counts lower than expected. Workflows and tips for troubleshooting both experiments are presented below. In general, it is easiest to troubleshoot in reverse order of the experimental protocol (start with data analysis and end with the array preparation steps).

The NP20 (sandwich method) and CM10 (retentate chromatography) experiments share many of the data analysis and acquisition steps. Many of the factors affecting array preparation are also shared between the two protocols (for example, pipetting errors and order of matrix and sample application). Since the CM10 experiment has more array preparation steps than the NP20 experiment, troubleshoot any “Fail” messages in the NP20 (sandwich method) experiment before proceeding to CM10 experiments.

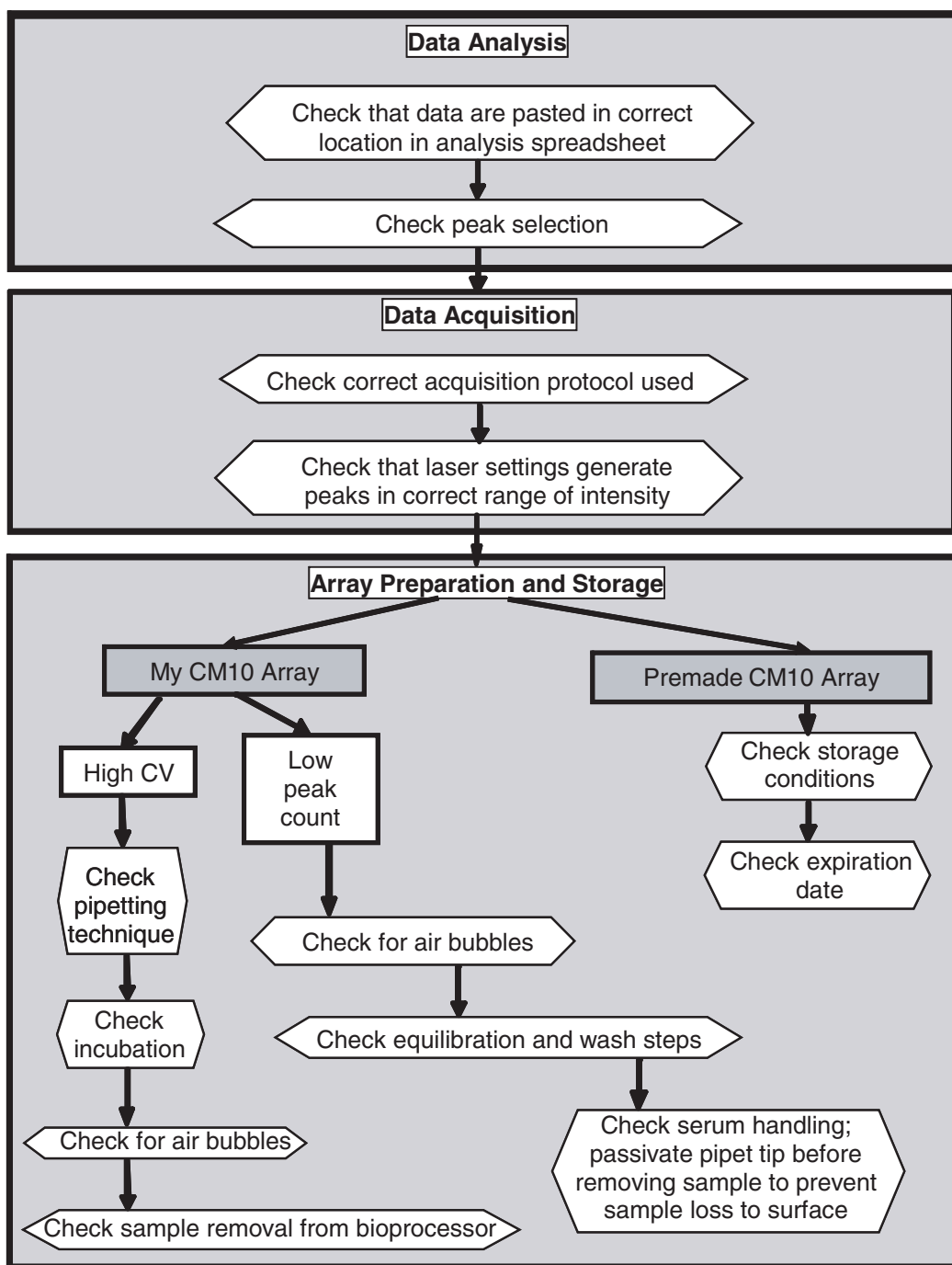
The most common sources for failing results. These errors can affect both the NP20 and CM10 experiments.

	Protocol Stage and Error	Symptom	Troubleshooting Advice
Data Analysis	Data pasted into incorrect cells of analysis spreadsheet	“Fail” message even if spectra look fine	Ensure that data are pasted to the correct cells in the spreadsheet
	Problems with peak selection	High CV	Check peak selection and repeat, if necessary zoom in to each peak before selection to ensure greatest accuracy
Data Acquisition	Incorrect data acquisition protocol	Low peak counts and/or high CV	Check the acquisition protocol. Repeat data acquisition from the arrays, if necessary (be sure to use new partitions)
	Laser settings are inappropriate	Low peak counts and/or high CV	Check that laser settings generate peaks in correct range of intensity Repeat optimization and data acquisition from the arrays, if necessary (be sure to use new partitions)
Array Preparation	Poor, inaccurate pipetting technique	High CV for both <i>E. coli</i> and serum	Examine raw data: if some spectra have significantly lower intensity within a sample type, sample may have been applied incorrectly Ensure consistent pipetting technique (smooth, continuous pipetting and accurate, calibrated, small-volume pipets appropriate to volume being transferred) Ensure pipetting technique is appropriate to sample type (for example, organic solvents must be transferred quickly to avoid leaking and evaporation, and viscous solutions require passivation of pipet tips)
	No sample added	Low peak counts and/or high CV	Examine raw data: not adding sample to one or more spots generates a spot with no peaks >500Da
	No matrix added	Low peak counts and/or high CV	Examine raw data: not adding matrix to one or more spots generates a spot with no peaks, especially notable in the region <500 Da (peaks generated by the matrix); peaks <500 Da indicate matrix was added effectively
	Sample and matrix applied in incorrect order (particularly in sandwich method)	Low peak counts	Examine raw data (refer to diagnostic figures below and repeat experiment) Check order of sample and matrix application
	Matrix crystallizes in pipet tip	High CV	Examine raw data: if some spectra have significantly lower overall intensity, sample may have been applied incorrectly Ensure no matrix crystals form on pipet tip
	Samples lost to pipet tip (more problematic with viscous samples, such as serum)	Low peak counts and/or high CV, especially if seen for serum only	Examine raw data: if some spectra have significantly lower intensity, sample may have been lost in transfer Passivate pipet tip (pipet up and down 3x before removing sample to prevent sample loss to surface)
	Old or contaminated matrix preparation used; incorrect mixing of matrix	Low signal and/or high CV; different peak ratios or higher or lower numbers of peaks detected	Should not be an issue for this kit, but it is a point to troubleshoot during experiments. Old or contaminated matrix, or matrix prepared incorrectly, can generate highly anomalous results
Array Storage (Pre-made Arrays)	Inappropriate storage conditions used	Low peak counts and/or high CV	Ensure arrays are stored at –20°C to –80°C; avoid exposing these arrays to vibration, moisture, or light
	Expiration date has been reached	Low peak counts and/or high CV	Use arrays before the expiration date Order new arrays (see Appendix D for ordering information)

Troubleshooting NP20 Array Analysis (Sandwich Method).



Troubleshooting CM10 Array Analysis (Retentate Chromatography Method).



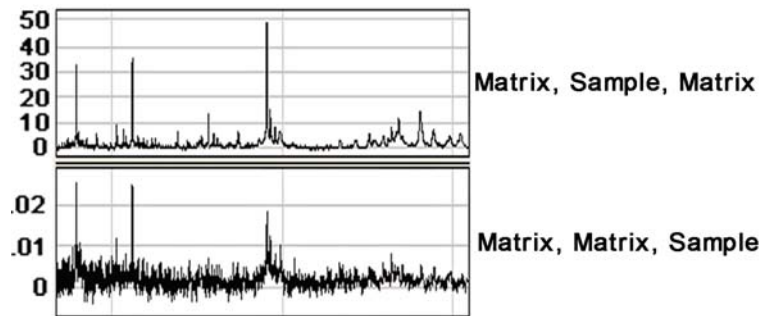
As mentioned above, many of the errors affecting the results of the NP20 experiment will also affect the CM10 experiment. The troubleshooting workflow for the CM10 experiment resembles that for the NP20 experiment, but with several additions related to steps specific to retentate chromatography.

Troubleshooting tips specific to retentate chromatography

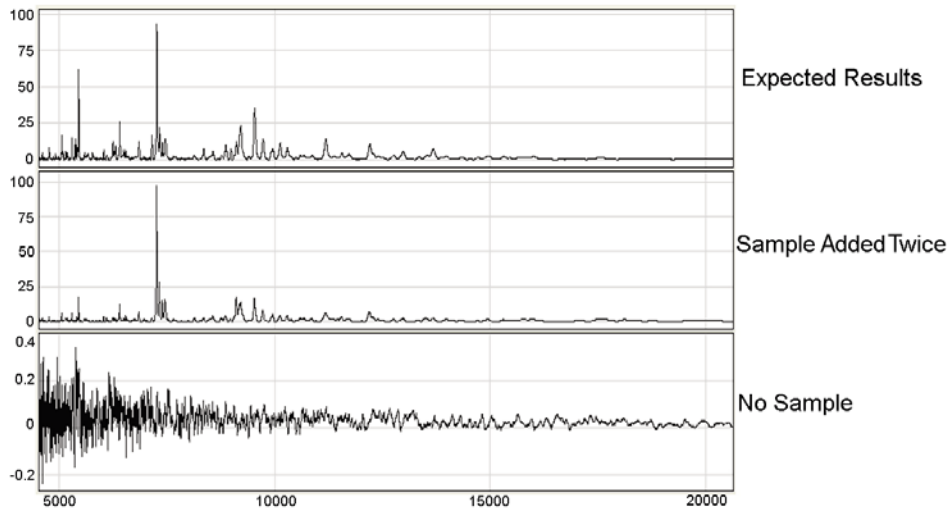
Symptom	Protocol Stage and Errors in Array Preparation	Troubleshooting Advice
Low signal	Equilibration step omitted or wrong reagents used	Ensure equilibration step is included and is performed using the correct buffer
High CV and low signal (partial bubble) or no signal (full bubble)	Bubbles in bioprocessor wells	Examine raw data: all spectra should be of comparable intensity; low intensity in any spectrum indicates bubble in bioprocessor, and no signal definitively identifies bubble
Low signal and/or high CV	Excess sample or buffer not removed completely from bioprocessor	Tap the bioprocessor vigorously against the paper towel to ensure all sample or buffer is completely removed before proceeding
Peak count may be slightly different (slight effect for the high-concentration samples used here)	Insufficient incubation Reagent or sample evaporation during incubation	Follow directions for proper incubation Seal the bioprocessor as directed during incubation
Low signal	Wash step(s) omitted or wrong reagent(s) used	Ensure both the buffer and wash steps are included and performed using the correct reagent(s)

Diagnostic Images

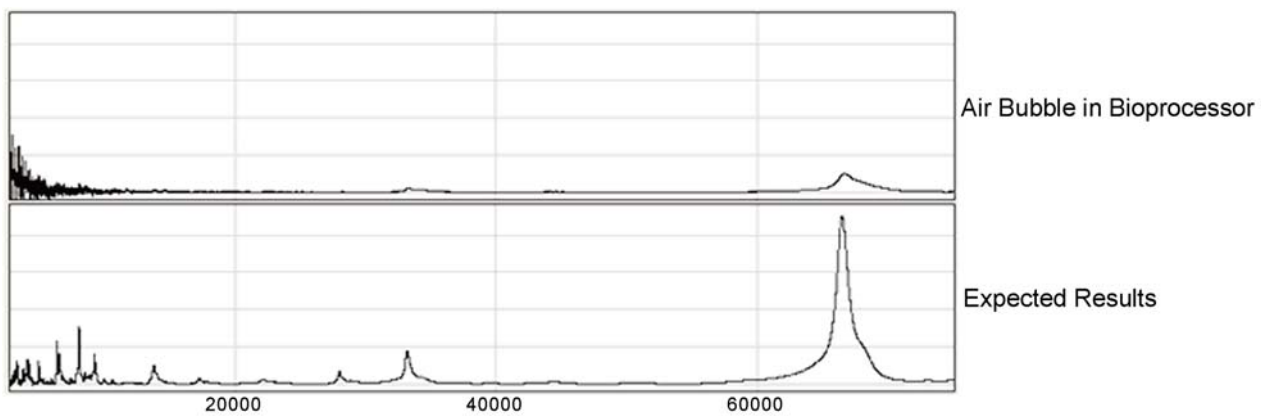
The order of sample and *E. coli* matrix addition in the sandwich method (MALDI) is critical. The top spectrum represents the expected profile from *E. coli*, the bottom spectrum shows an overall reduced peak intensity.



The spectra below illustrate the effects of sample addition errors. Adding twice the sample (middle figure) causes sample saturation that can result in variation in the profile. No sample results in only matrix peaks.



An air bubble in the bioprocessor can limit the contact of the sample with the surface of the array resulting in spectra with either no or very small peaks.



Appendix D

Ordering Information

C70-00068	ProteinChip® Starter Kit , includes 3 ProteinChip CM10 arrays, 3 ProteinChip NP20 arrays, 1 ProteinChip premade CM10 array, 1 Proteinchip premade NP20 array, 1 bottle CM low-stringency buffer, 2 vials SPA, 1 vial 1% TFA, 1 vial ACN, 1 peptide standard array, 12 blank arrays, 2 cassettes, 1 bioprocessor reservoir, instruction manual, and CD with protocols, instructional video, and data analysis spreadsheet
C70-00070	ProteinChip Peptide Mass Calibration Kit , includes 1 peptide standard array and instructions
C70-00080	ProteinChip OQ Kit , includes 2 detector calibration arrays, 6 detector qualification arrays, 2 peptide standard arrays, CD with protocols, instructions
C70-00081	ProteinChip System Check Kit , includes 1 detector calibration array, 1 detector qualification array, 1 peptide standard array, CD with protocols, instruction manual
C50-30011	ProteinChip Cassette-Compatible Bioprocessor , includes ProteinChip array forceps, cassette hold-down frame, 12 blank ProteinChip arrays
C50-30013	ProteinChip Cassettes , empty, hold 12 ProteinChip arrays, 5
C57-30075	ProteinChip CM10 Arrays , A–H format, 12
C57-30043	ProteinChip NP20 Arrays , A–H format, 12
C30-00002	ProteinChip SPA Matrix , 5 mg/vial, 20
C10-00006	ProteinChip Human Serum Control , 500 µl
163-2111	<i>E. coli</i> Protein Sample , lyophilized, 2.7 mg
P20-10001	ProteinChip SELDI System Basic Course



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Canada 905 364 3435 **China** 86 21 6426 0808 **Czech Republic** 420 241 430 532 **Denmark** 44 52 10 00 **Finland** 09 804 22 00 **France** 01 47 95 69 65
Germany 089 318 84 0 **Greece** 30 210 777 4396 **Hong Kong** 852 2789 3300 **Hungary** 36 1 455 8800 **India** 91 124 4029300 **Israel** 03 963 6050
Italy 39 02 216091 **Japan** 03 6361 7000 **Korea** 82 2 3473 4460 **Mexico** 52 555 488 7670 **The Netherlands** 0318 540666 **New Zealand** 0508 805 500
Norway 23 38 41 30 **Poland** 48 22 331 99 99 **Portugal** 351 21 472 7700 **Russia** 7 495 721 14 04 **Singapore** 65 6415 3188 **South Africa** 27 861 246 723
Spain 34 91 590 5200 **Sweden** 08 555 12700 **Switzerland** 061 717 95 55 **Taiwan** 886 2 2578 7189 **United Kingdom** 020 8328 2000

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